Chapter 6. Volume and Open Interest

Lecture notes | First semester

Benjamin C. Sotelo | Institute of Accounts, Business and Finance, FEU Manila

Start here

What this chapter is about

These notes are the record of what Chapter 6 covered, written to be read on their own. If you were in the room, they are what to revise from. If you missed the session, they are the session. They follow the lecture in the same order. Sections 1 to 5 are the book's long section 6.1, in its own order, and Section 6 is its section 6.2; the book's 6.3, its summary, is the summary at the back.

The chapter reads volume, and then open interest, against price. Read it with the pictures beside you. A figure is the book's. A chart is one this course drew, on invented prices, where the book makes a point in words, in a sketch, or on a real chart too dense to read the point off. Every term is defined once, in the same words as the slides, and listed again at the back. The check yourself boxes are not assessed.

The chapter leans on tools a later chapter of the book teaches, oscillators above all, and in a few places it says two different things. Where that happens these notes say so rather than filling the gap from somewhere else. One of the book's six learning objectives, reading volume across successive contract months in futures, is not taught in this chapter's text; Chapter 3 covered it.

After studying this chapter you should be able to:

1. Understand volume action and its implications for market action.
2. Recognize early trend weakness via volume and open interest divergence.
3. Identify bullish and bearish signals on volume and open interest indicators.
4. Forecast potential reversals and corrections via volume cycles.
5. Set up effective volume filtering, for better timed entries and exits.
6. Read volume across successive contract months in futures markets.

Chapter scope follows Lim, M. (2016), The Handbook of Technical Analysis, chapter 6. Figures are reproduced from that text and remain the publisher's copyright. Charts are drawn for this course on invented prices.

Section 1

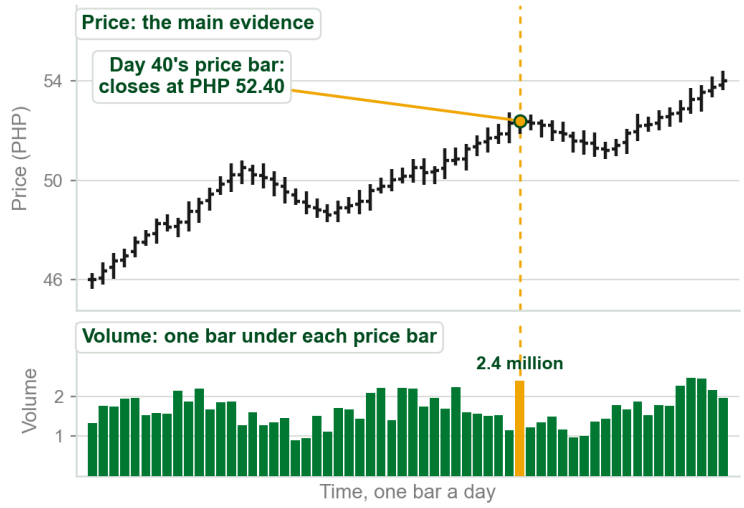
Confirmation and Divergence

What volume is, when it is bullish and when bearish, what volume divergence and confirmation mean, how they differ from the divergence of Chapters 4 and 5, and the two kinds of volume oscillator. The book's section 6.1, pages 173 to 180.

1.1 Price first, volume as evidence

The chapter opens by putting volume in its place. Price is regarded as the best indicator of potential future price behavior. Volume and open interest offer additional evidence to corroborate a bullish or bearish setup, can show underlying weakness, and can work as timing indicators for potential breakouts.

The book also warns that many novices misread declining volume as a weakness in a consolidation; Section 2 explains why that is a misreading.



Invented prices and volume. Volume in millions of shares.

Chart A Price bars over their volume, one day picked out.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart A shows how every figure in this chapter is drawn. The price bars are in the upper panel and the volume bars in the lower panel, one volume bar standing directly under each price bar. Find the gold day: its price bar says where it traded and closed, and the bar under it says how much traded.

1.2 Chapter 2's tenet, recalled

Chapter 2 taught Dow's sixth tenet: volume must confirm the trend. Volume should expand in the direction of the preexisting trend. In an uptrend it should rise as price rises and fall in downside retracements; the book calls that bullish volume action. In a downtrend it should rise as price falls and fall in upside retracements: bearish volume action. Volume is a supporting, or secondary, indicator. Any volume action contrary to this is an indication that an uptrend may potentially be bearish and a downtrend bullish.

Two sketches of volume confirming a trend. Top: a rising price with volume bars that grow on the rise and shrink in the downside retracement, labelled bullish volume action in an uptrend. Bottom: a falling price with volume that grows on the fall and shrinks in the upside retracement, labelled bearish volume action in a downtrend.

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Figure 6.1 Volume confirming a preexisting trend.

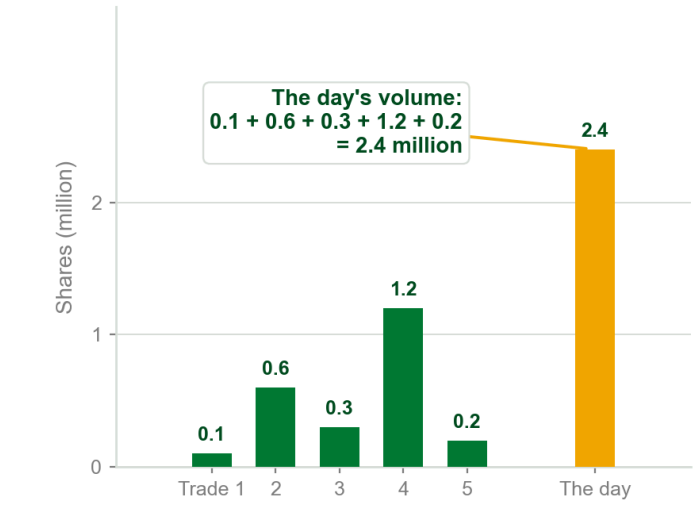
Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 6.1 draws both halves of the tenet. In each sketch, look for where the tall volume bars sit: under the move that goes with the trend. The short ones sit under the retracement.

1.3 Volume, and the two principles

Volume has been on our charts since Chapter 1, and this is where the book defines it. Suppose one day sees five trades, of 0.1, 0.6, 0.3, 1.2 and 0.2 million shares. That day's volume is their total, 2.4 million shares. For a futures market the count is in contracts.

> **Volume.** Volume is an indication of market participation. It is the total number of shares or contracts traded over a specific time interval.



Our example: one day's five trades, in million shares.

Chart B One day's five trades, added up to its volume.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart B sets out that sum: five columns, one per trade, and the day's total in gold at the right.

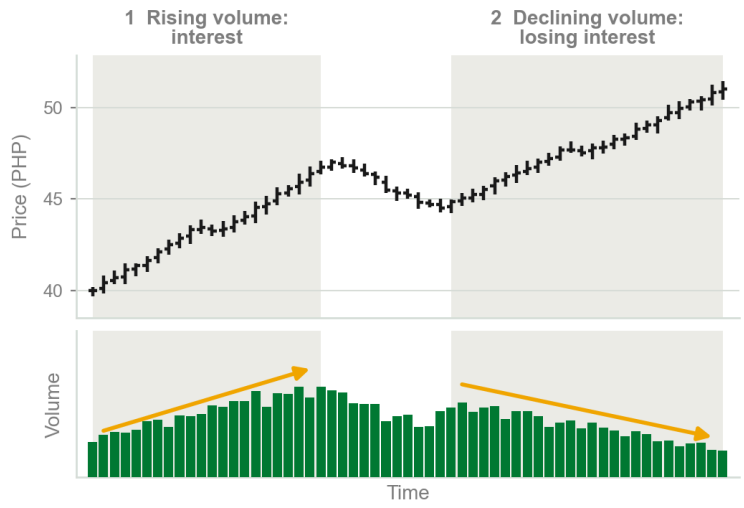
The book asks for two principles to be remembered whenever volume is studied. First, a rise in volume means participants want to see price go higher in an uptrend, or lower in a downtrend, and are willing to buy higher or sell lower to take part. Second, a decline in volume means they are losing interest, and would rather buy lower in an uptrend or sell higher in a downtrend, if they are not exiting altogether. **Volume, in short, measures participation.**

In Chart C both rallies rise. Read the gold arrows in the volume panel: under the first rally, numbered 1, volume grows as buyers join; under the second, numbered 2, it shrinks as they lose interest.

1.4 Bullish volume and bearish volume

The book states the next rule four times, once for each trend and each direction of volume, and all four say the same thing. Volume that moves with price is bullish; volume that moves against price is bearish. A share that rallies on rising volume and dips on falling volume shows bullish volume both times. A share that falls on rising volume and bounces on falling volume shows bearish volume both times.

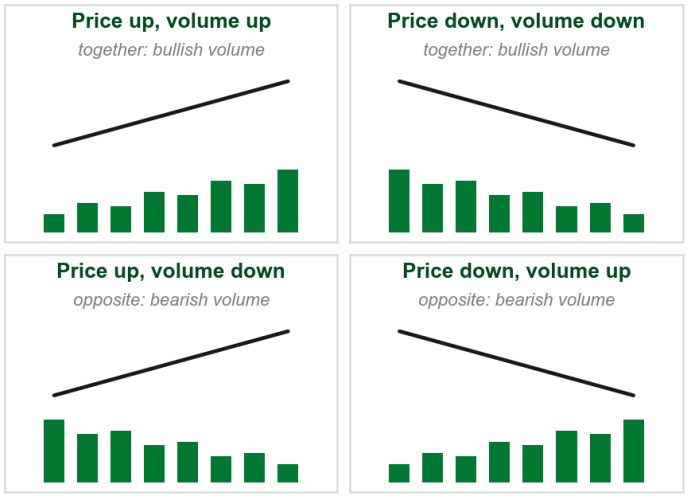
> **Bullish volume and bearish volume.** Both rising and declining volume are regarded as bullish volume if volume and price rise and decline together, and as bearish volume if they move in opposite directions.



Invented prices and volume. Both rallies rise; only the volume under them differs.

Chart C Two rallies: one on rising volume, one on falling.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.



Not prices: sketches. In each case a price line over its volume.

Chart D The four cases of price and volume together.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart D puts the four cases in a grid. The top row is volume moving with price, bullish; the bottom row is volume moving against it, bearish. Notice that falling volume appears in both rows: on its own it means nothing until the direction of price is known.

That is the point of the book's warning. Declining volume does not necessarily imply bearishness, and rising volume does not imply bullishness. The book calls the opposite reading a simple mistake that many novices make.

Figure 6.2 is the continuous daily chart of gold. Look at the volume panel through the long rise, expanding with the trend, and then at the retracement at the right, where price and volume fall together. That fall in volume is bullish.

Figure 6.3 swaps the volume of Figure 6.1. In an uptrend whose volume shrinks on the rises and grows in the retracements, the volume is bearish: a potentially weak or bearish uptrend.

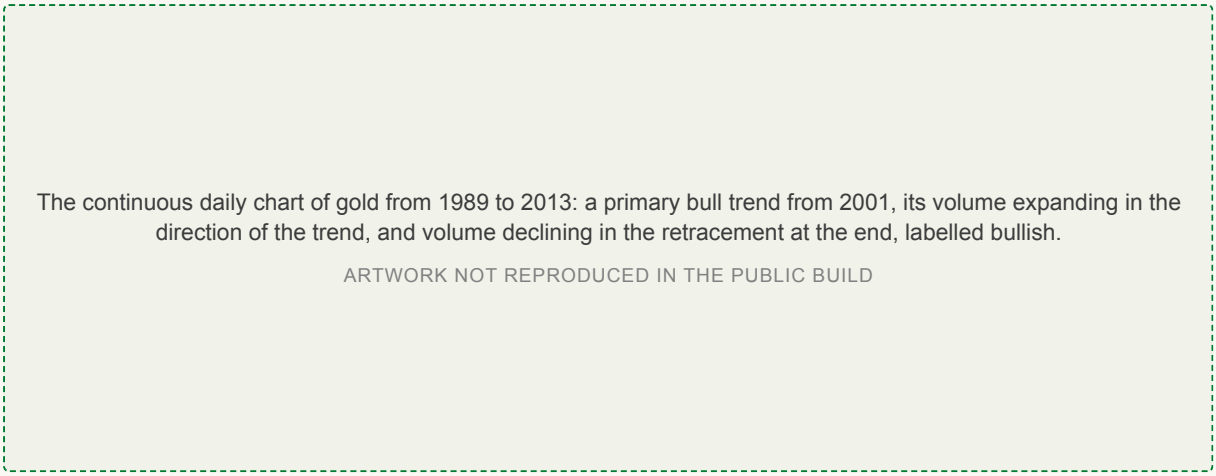


Figure 6.2 Bullish volume in gold's primary bull trend.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)



Figure 6.3 Volume not confirming a preexisting trend.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

In a downtrend whose volume shrinks on the falls and grows in the upside retracements, it is bullish: a potentially bullish downtrend.

CHECK YOURSELF

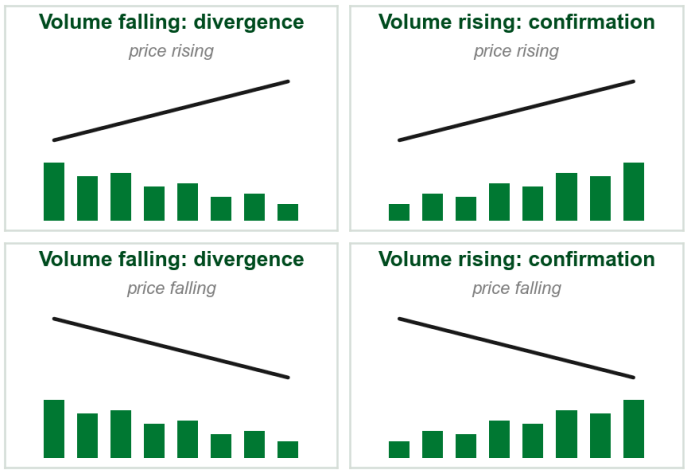
In an uptrend, price dips for a week and volume falls during the dip. Is that volume bullish or bearish, and why does the fall in volume not settle it on its own?

1.5 Volume divergence and confirmation

The book now introduces a second pair of words, and this pair ignores price altogether. Volume falling is divergence. Volume rising is confirmation, which the book also calls convergence. If a share rises from 40 to 46 pesos while its daily volume shrinks from 3 million to 1 million shares, that is volume divergence; had volume grown to 5 million, it would be confirmation.

> **Volume divergence and volume confirmation.** Volume divergence occurs whenever volume declines, irrespective of the direction of price. Volume convergence or confirmation occurs whenever volume rises, irrespective of the direction of price.

Read Chart E by its columns. The left column is volume falling, divergence, once under a rising price and once under a falling one. The right column is volume rising, confirmation, under both. Price goes both ways in each column.

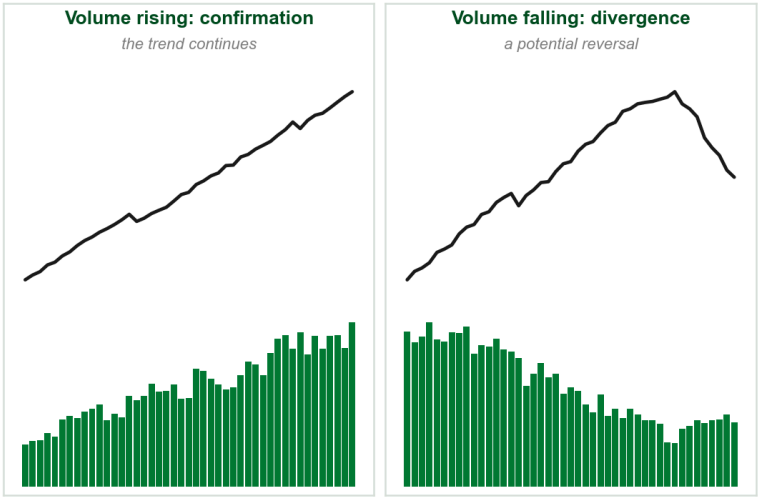


Not prices: sketches. Left column: volume falls. Right: it rises. Price goes both ways in each column.

Chart E Divergence and confirmation, each under both kinds of price move.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Divergence says participants are not interested in seeing price go further: bearish for an uptrend, bullish for a downtrend. Confirmation says they are interested in taking part: bullish for an uptrend, bearish for a downtrend. **Confirmation is an early indication of a potential continuation of the trend, and divergence of a potential reversal.** The book adds that divergence always indicates potential weakness in an existing trend and confirmation always potential strength.



Invented prices and volume. The right-hand ending is one possibility, drawn to show the warning.

Chart F Confirmation continuing, divergence turning.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart F shows the two side by side. In the left uptrend volume rises with price and the trend continues. In the right one volume falls as price rises, and price turns down at the end. That ending is one possibility, drawn to show what the warning means; divergence warns and does not guarantee.

1.6 Four names

Put the two pairs together and there are four names. Price and volume falling together is bullish divergence. Both rising is bullish confirmation. Volume falling while price rises is bearish divergence, and volume rising while price falls is bearish confirmation. That list is the book's own summary of every scenario.

> **Bullish and bearish divergence and confirmation.** Bullish divergence can only occur in a downtrend, a potential upside reversal; bearish divergence only in an uptrend, a potential downside reversal.

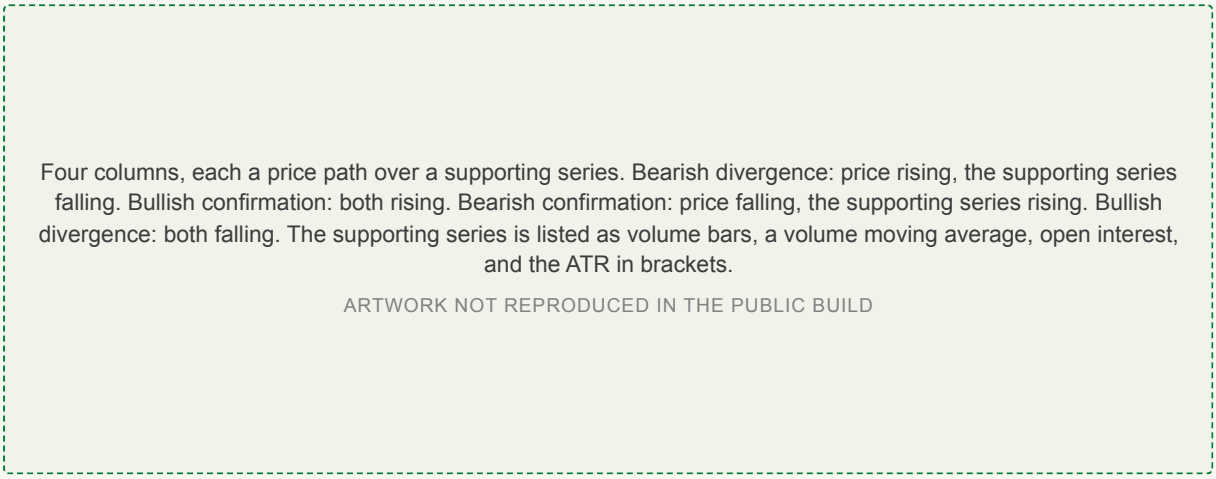


Figure 6.4 The four names, as price over a supporting series.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

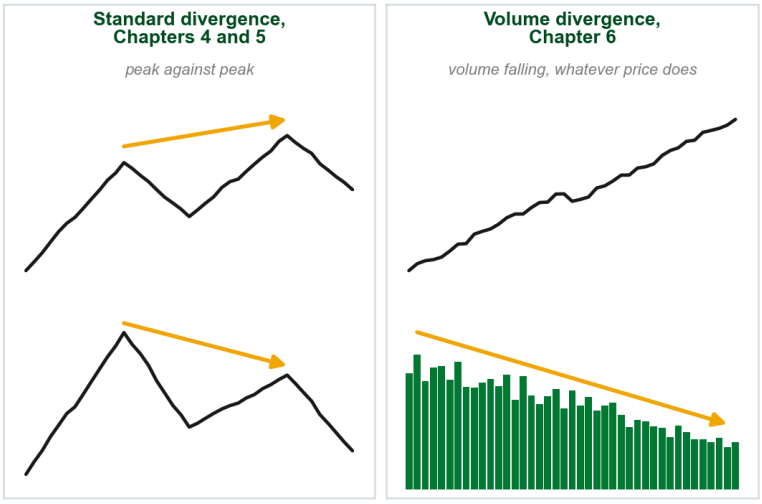
Figure 6.4 draws the four as columns, each a price path over a supporting series. The book's figure lists volume bars, a volume moving average, open interest, and, in brackets, the ATR as supporting series. The ATR appears in no sentence of the chapter's text.

One place where the book says two things. It says bullish divergence can only occur in a downtrend, yet Figure 6.6 below finds bullish divergence in two dips inside an uptrend. These notes read downtrend there as price falling, for any stretch, which is how the summary list reads it. That reading is ours.

1.7 Not the divergence of Chapters 4 and 5

Chapters 4 and 5 showed standard divergence: price makes a higher peak while an oscillator, the MACD in Chapter 4 or the averaged bar stochastic in Chapter 5, makes a lower one. That test compares two peaks on two lines. Volume divergence does not. **Unlike standard price-based oscillators, the direction of price is irrelevant to whether volume diverges.** Price's direction is still needed to say whether the divergence is bullish or bearish. The book also says there is no reverse divergence classification for volume, and sends the classification of divergence in general to its Chapter 9. This is review question 3.

Chart G puts the two tests side by side. On the left, follow the two gold arrows: price's second peak is higher, the oscillator's lower. On the right there are no peaks to compare: the price rises and the volume under it falls, and that alone is volume divergence.



Invented prices and volume. Left: price over an oscillator. Right: price over its volume.

Chart G Standard divergence beside volume divergence.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

1.8 Two worked charts

An intraday chart of Isis Pharmaceuticals: price falls steeply and curves into a larger rounding bottom pattern, and the volume panel under it carries two dashed lines falling to the right, each labelled bullish volume divergence.

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Figure 6.5 Bullish volume divergence on Isis Pharmaceuticals.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 6.5 is Isis Pharmaceuticals tracing out a bullish rounding bottom, a pattern Chapter 4 named. Look at the two dashed lines in the volume panel, each labelled bullish volume divergence: price falls and volume falls with it. The book calls it divergence providing additional evidence of a bullish chart formation.

Figure 6.6 splits the SPDR Gold Trust into ten periods; its caption says daily, though the chart is labelled weekly. Periods 1, 3, 7 and 9 show bullish confirmation, 4 and 6 bullish divergence, and 5 flat volume. In periods 2, 8 and 10 volume declines during a consolidation, where it works as a breakout timing indicator: at the end of each, price breaks out as volume reaches its minimum for the period.

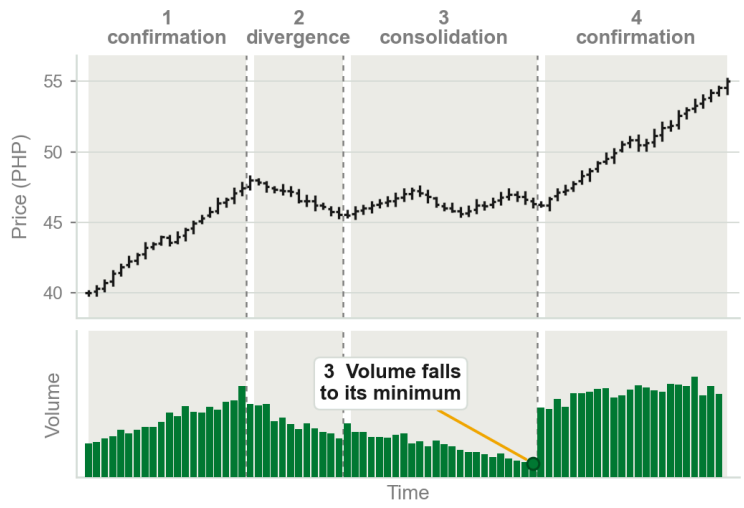
Chart H is the same reading on one invented uptrend, so each kind of period can be seen. Period 1 rises on rising volume, bullish confirmation. Period 2 dips on falling volume, bullish divergence. Period 3 moves sideways while volume falls to its minimum, a consolidation. Period 4 breaks out and rises on rising volume. Every period is bullish volume.

The SPDR Gold Trust over about four years, split by dashed lines into ten numbered periods, with arrows on the volume panel showing volume rising or falling in each, under the label volume expanding in the direction of the existing trend.

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Figure 6.6 Ten periods of bullish volume in the SPDR Gold Trust.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)



Invented prices and volume. Every period is bullish volume: price and volume move together.

Chart H One invented uptrend in four periods.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

CHECK YOURSELF

Price makes a higher high while volume makes a lower high. Is that volume divergence? Would it still be volume divergence if price had made a lower high instead?

1.9 Two kinds of volume oscillator

The book calls volume indicators drawn in their own panel under price volume oscillators, and sorts them into two kinds for the purpose of reading divergence. Plain volume bars count shares and nothing else. A line that adds the day's volume when the close rises and subtracts it when the close falls uses price as well.

> **Price-based and non-price-based volume oscillators.** Price-based volume oscillators incorporate price data in their construction, whereas non-price-based volume oscillators only incorporate volume data in their construction.

Chart I draws three panels over the same days: the closes, the volume bars, which are non-price-based, and a line that needs the closes to be drawn, which is price-based.



Invented prices and volume. Bottom line: add the day's volume on a higher close, subtract it on a lower one. Part 6 names it the OBV.

Chart I Price, volume bars, and a price-based line from the same days.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

That bottom line is the On Balance Volume, taught in Section 6.

The book's two lists

- Price-based: On Balance Volume (OBV), the Accumulation/Distribution Line (ADL), Chaikin Money Flow, the Chaikin Oscillator, the Demand Index and the Herrick Payoff Index.
- Non-price-based: standard volume bars, smoothed volume, detrended volume, the rate of change of volume (ROC volume) and the percentage volume oscillator (PVO).

The OBV, the ADL, the two Chaikin indicators and the Herrick index are taught in Section 6 and smoothed volume in Section 3. The Demand Index, detrended volume, ROC volume and the PVO are named in the chapter and never explained.

Eight pairs of arrows, a price arrow over a volume arrow, in four boxes: divergence and confirmation, each for non-price-based and price-based volume oscillators. The two kinds differ only in bullish divergence and bearish confirmation, where the price-based volume arrow points the other way.

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Figure 6.7 Reading divergence on the two kinds.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 6.7 is eight pairs of arrows. Compare the boxes left with right. Bearish divergence and bullish confirmation are read the same way on both kinds. The two differ in bullish divergence, where price falls and a non-price-based oscillator falls with it but a price-based one rises, and in bearish confirmation, where price falls and a non-price-based oscillator rises but a price-based one falls.

In short, a price-based oscillator diverges by moving against price, and plain volume by falling. That one-line summary is ours.

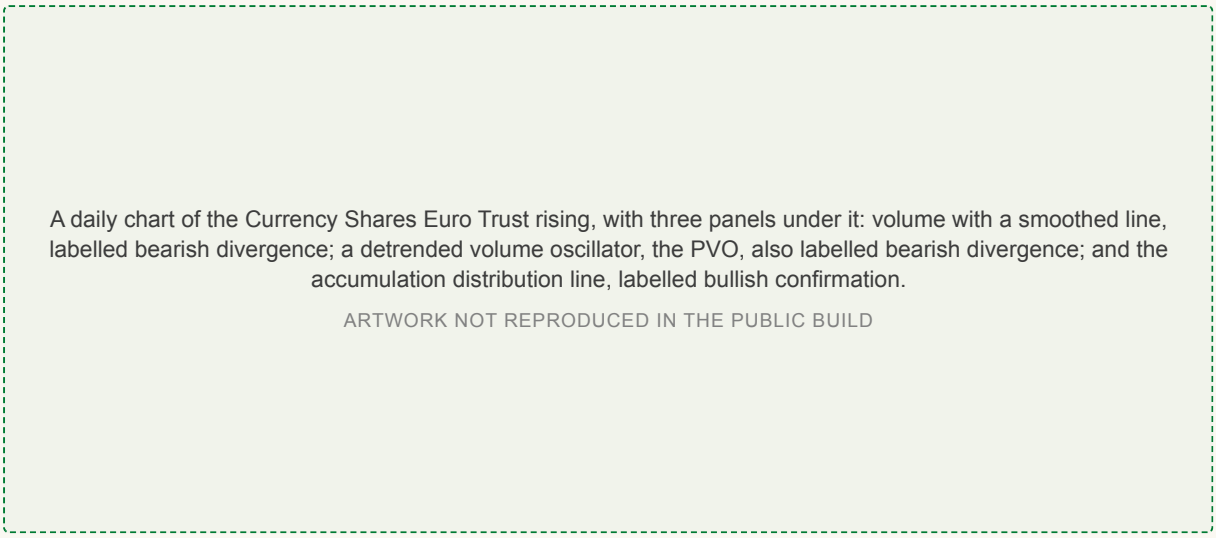


Figure 6.8 Both kinds read on the Currency Shares Euro Trust.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 6.8 reads one rise both ways. The volume with its smoothed line and the detrended PVO, both non-price-based, show bearish divergence. The ADL, price-based, shows bullish confirmation. Each kind is read by its own rule; the book does not say which to believe.

Section 2

Timing, the VWMA and Climaxes

Why minimum volume times a breakout or a reversal, the volume-weighted moving average and its two readings, the four kinds of top and bottom, where to set an overextension level, and how the timeframe changes what volume shows. The book's section 6.1, pages 180 to 186.

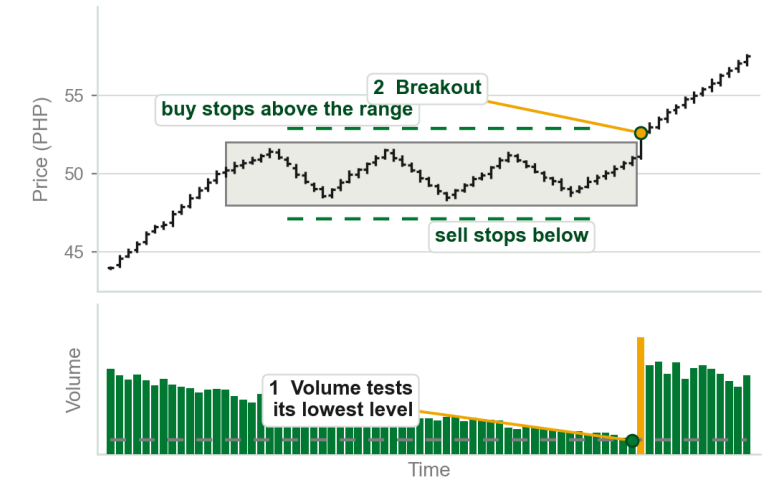
2.1 Volume as a timing indicator

Inside a consolidation, whenever volume declines close to its lowest historical level, a breakout is expected. The book's reason is that by then most buyers who wanted to buy have bought and most sellers who wanted to sell have sold. Stop orders, taught in Chapters 1 and 5, accumulate above and below the range over the consolidation, and once triggered they are the catalyst for the breakout, up or down. Breakout here is the use of Chapters 2 and 4: price leaving a range.

This answers the novice error of Section 1. **There is no divergence reading during a consolidation, because price is moving sideways.** Falling volume inside a range is not weakness; it is the clock running down to the breakout.

In Chart J find the box, the range, with buy stops drawn above it and sell stops below. Then follow the volume panel to point 1, where volume tests its lowest level, and up to point 2, where price breaks out of the top of the box and volume surges. Where the stops sit is our drawing of the book's sentence.

> **Low volume breakout signal.** Whenever the volume is testing a significant minimum or its lowest historical levels, a breakout is expected. We refer to this as a low volume breakout signal.



Invented prices and volume. Dashed: volume's lowest level.
The book says a breakout may be up or down; this one is up.

Chart J A range, its stops, and volume falling to its lowest level.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

A daily Apple chart with a low volume level drawn across its volume panel. Circles mark two low volume reversals at time lines 1 and 2, a bearish divergence and a bearish confirmation, and later an accumulation box that price leaves on a low volume breakout signal.

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Figure 6.9 Volume as a reversal and breakout timing indicator on Apple.

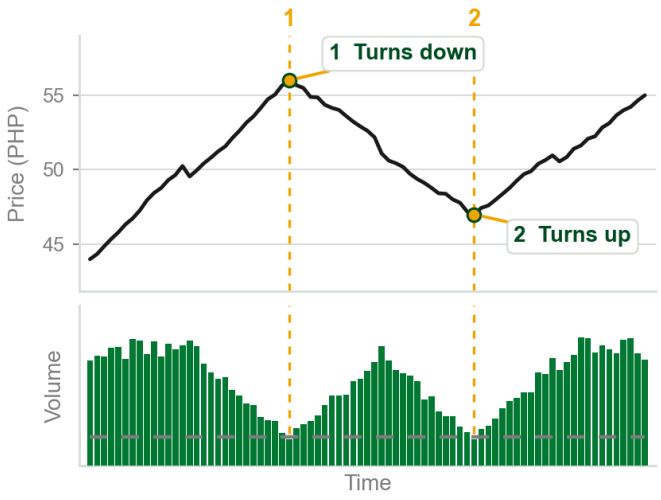
Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 6.9 is a daily Apple chart. Find the box at the bottom right of the price panel, an accumulation, and the circle under it where volume sits on its low volume level: price then breaks out of the box. The same figure also shows two turns on low volume, at time lines 1 and 2, before the consolidation.

Those turns are the second use of minimum volume. The book says there is a tendency for price to reverse when volume is at its minimum or lowest historical levels.

> **Low volume reversal.** There is a tendency for price to reverse when volume is at its minimum or lowest historical levels. We refer to such reversals as low volume reversals.

Chart K draws it on one invented line. At 1 price peaks and at 2 it bottoms, and at both the volume bar under the turn touches the dashed low volume level. The book gives no rule for where that level goes; Figure 6.9 draws it by eye, and so does the chart.



Invented prices and volume. Dashed: the low volume level, drawn by eye.

Chart K A low volume reversal at a peak and at a trough.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

The daily iShares Silver Trust with three consolidations boxed and a low volume threshold drawn across the volume panel. Dotted arrows show volume falling to the threshold in each box, and the breakouts at bars 1, 2 and 3 follow, with volume rising on each breakout bar.

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Figure 6.10 Three breakouts at minimum volume in silver.

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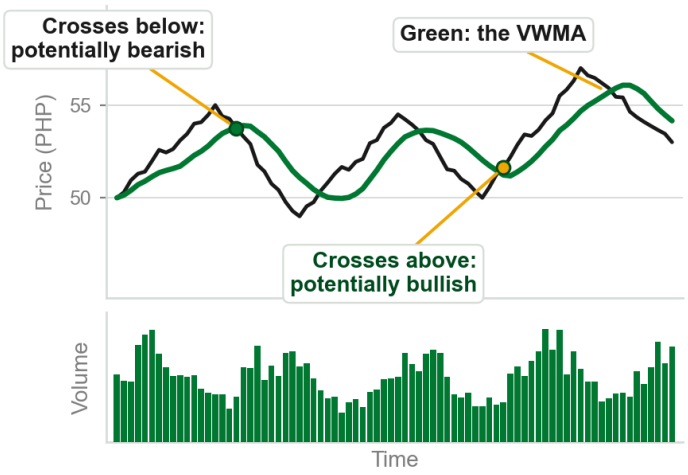
Figure 6.10 shows the breakout signal three times on the iShares Silver Trust: one accumulation breakout up and two distribution breakouts down, at bars 1, 2 and 3, each as volume tests the threshold. Notice that the breakout bars themselves carry more volume than the bars before them: volume falls into the breakout and surges on it.

2.2 The volume-weighted moving average

The book introduces a volume-weighted moving average, VWMA, as a line of average price on the chart, weighted by volume as its name says. It defers the calculation to its Chapter 11, and moving averages themselves, met in Chapters 4 and 5, are taught there. For now the line matters by which side of it price is on: a share at 52 pesos above a VWMA at 50 is potentially bullish, and at 48, below it, potentially bearish.

> **Volume-weighted moving average, VWMA.** A VWMA may be created to track overvalued and undervalued stock. Typically, price is considered potentially bullish when it is above the VWMA, and bearish when below it.

Chart L draws an invented price and a 10-bar VWMA we computed.



Invented. Green: a 10-bar VWMA we computed.
The book's method is in its Chapter 11.

Chart L A price and its 10-bar VWMA, with two crossings.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Find the two labelled crossings: below the line, potentially bearish; above it, potentially bullish.

A 10-minute chart of the S&P 500 SPDR over four days with a dashed volume weighted average price line, labelled price above it bullish and below it bearish, and the line marked as support and as resistance. Circles on the volume panel mark the daily volume cycle peaks, with arrows from them to the crossovers.

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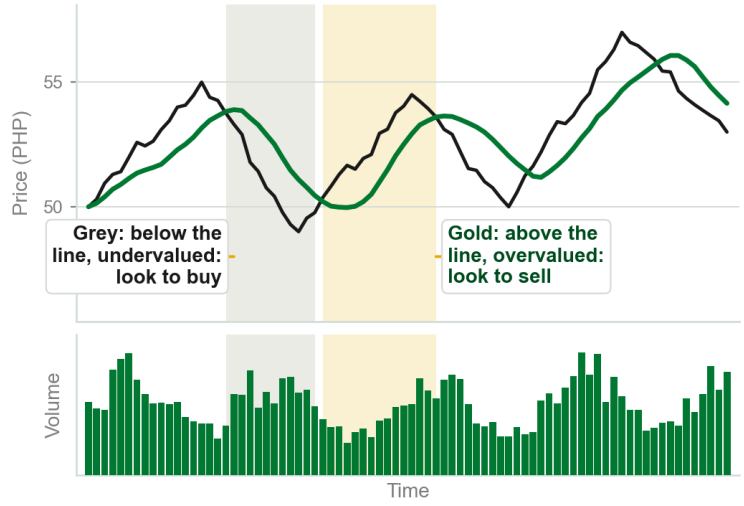
Figure 6.11 The VWMA on the S&P 500 SPDR.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 6.11 shows price rising decisively every time it crosses above the line and declining when it crosses below. Follow the dashed arrows from the circled volume peaks to the crossovers: the book notes that the crossovers occur at the daily volume cycle peaks, which Section 5 explains. The figure also marks the line acting as support and as resistance. Its caption says VWMA on a daily chart; the figure itself is labelled VWAP and is a 10-minute chart.

A paragraph later the book reads the same line a second way. A stock below the VWMA is undervalued and above it overvalued, so many institutional traders look for buy signals only when it is below and sell signals only when it is above. **So above the line is potentially bullish and also where institutions look to sell; the book gives both readings and does not reconcile them.**

Chart M shades one stretch below the VWMA, undervalued, where institutions look to buy, and one above it, overvalued, where they look to sell.



Invented. Green: a 10-bar VWMA we computed. Shaded: the stretches the labels name.

Chart M Undervalued below the line, overvalued above it.

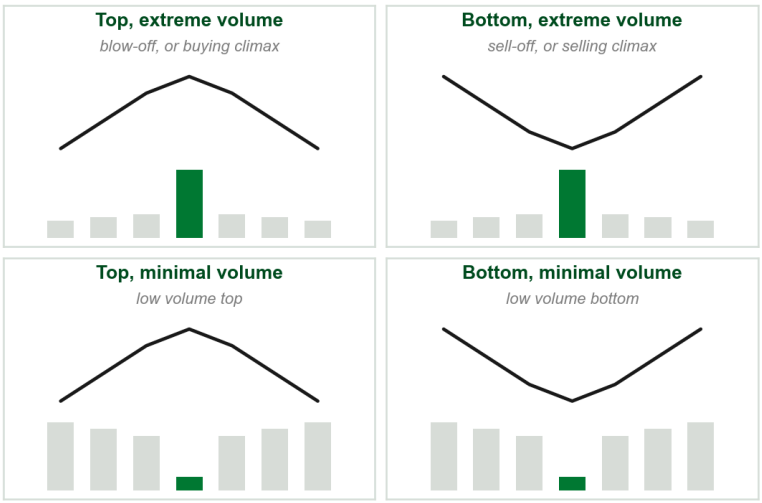
FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

CHECK YOURSELF

Volume in a range has fallen to its lowest level in a year. What does the book expect next, and can it tell you which way?

2.3 Climaxes, and the overextended levels

A market top or bottom is usually accompanied by relatively large or extreme volume, or by relatively low or minimal volume. That gives four cases. A top on extreme volume is a blow-off or buying climax, and a bottom on extreme volume a sell-off or selling climax. Chapter 4 named the climaxes that end a trend by who buys and who sells in them; Chapter 6 adds their extreme volume. A top on extremely low volume is a low volume top, and a bottom on it a low volume bottom.



Not prices: sketches. The green bar is the volume on the day of the turn.

Chart N Four turns, named by place and by volume.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart N sketches the four. In each, the green bar is the volume on the day of the turn: towering in the top row, the climaxes, and the smallest of all in the bottom row.

To use extreme volume as a signal, the book draws a line across the volume panel at the height that only extreme volume reaches. A lower line catches smaller turns.

> **Upper and lower overextended volume levels.** The extreme volume level is the upper overextended volume level, a signal for potential bottoms or tops. A lower overextended volume level forecasts less significant corrections.

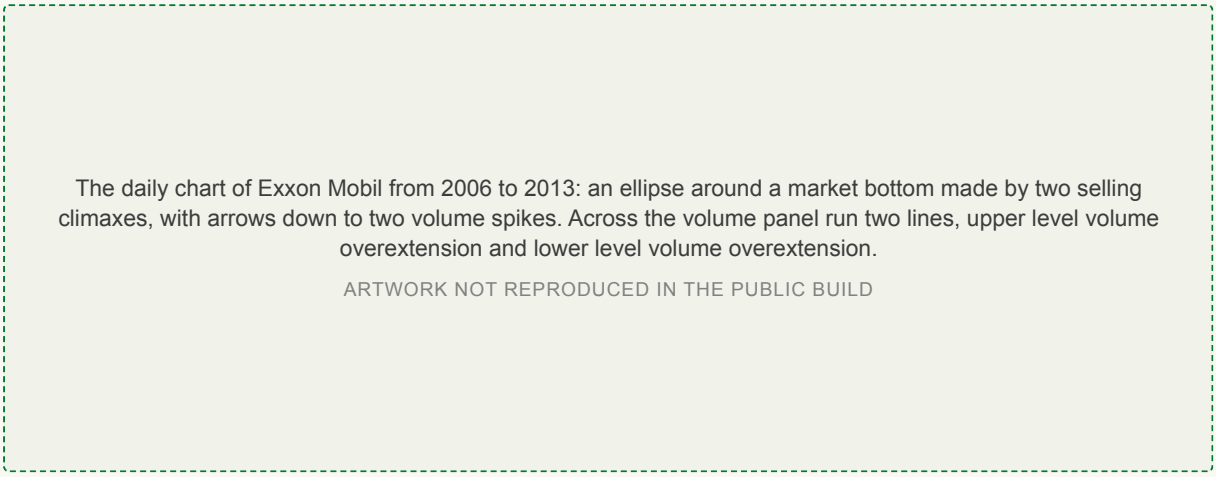


Figure 6.12 Selling climaxes at the upper overextended level, Exxon Mobil.
Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 6.12 is the daily chart of Exxon Mobil, which the book spells Exxon Mobile. Look at the ellipse around the market bottom and follow its arrows down to two volume spikes, both of which reach the upper line: two selling climaxes.

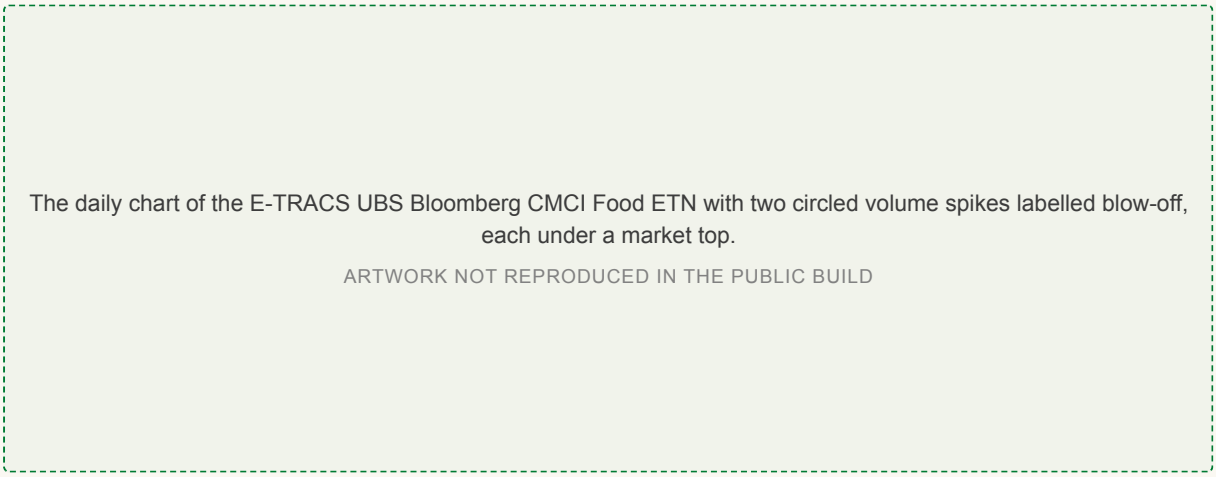


Figure 6.13 Blow-offs on the E-TRACS CMCI Food ETN.
Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 6.13 is the mirror case: a couple of blow-offs, or buying climaxes, at relatively high volume. Follow each BLOW-OFF arrow to its circled spike and up to its top.

Where should the line go? The book's answer is by visual inspection, at a level where only the most significant tops or bottoms are found. **Set lower, it signals more often, and the tops and bottoms it signals are less significant.** In Figure 6.14 the overextended spikes are all above the rolling 100-day 2-sigma level, a computed line that Section 3 builds, and at the right price breaks a significant support on extreme volume.

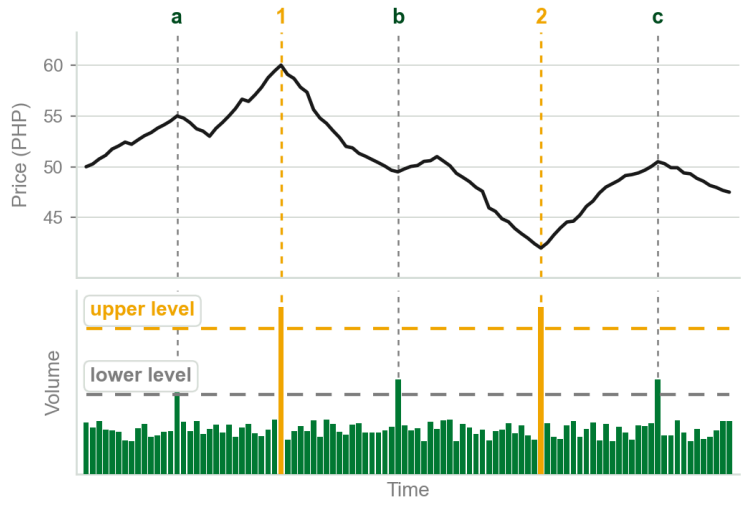
Chart O draws both lines on one invented chart. Only the spikes at the two major turns, numbered 1 and 2, reach the upper line.

The daily SPDR Gold Trust: a dashed volume overextension line across the volume panel with circled spikes above it, and a dashed support line on the price that is broken at the right on extreme volume, labelled above the rolling 100-day 2-sigma level.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 6.14 Volume overextensions in the SPDR Gold Trust.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)



Invented. 1 and 2: the major turns. a to c: smaller corrections. Levels drawn by eye.

Chart O A higher and a lower overextension level on one chart.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Three more, at smaller corrections lettered a to c, reach only the lower one. Both lines are drawn by eye.

The continuous daily chart of silver with an upper and a lower volume overextension line, and under it a bandwidth oscillator labelled volatility, with its own historical volatility level. Thin lines join circled volatility spikes to volume overextensions and to a top at 1 and a bottom at 2.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 6.15 Volatility spikes at the same turns in silver.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 6.15 adds a second kind of evidence. A Bollinger Bandwidth oscillator times volatility in silver, and its spikes coincide with the volume overextensions at the top, 1, and the bottom, 2. Many practitioners filter out volatility below a threshold and regard it as noise. The Bollinger Band and its Bandwidth are named and not explained in this chapter.

2.4 Low volume tops and bottoms

Tops and bottoms can also form on very little volume. These are the low volume reversals of Section 2.1, now named by where they turn.

> **Low volume top and low volume bottom.** A market top associated with extremely low or minimal volume is called a low volume top. A market bottom associated with extremely low volume is called a low volume bottom.



Figure 6.16 Low volume reversals on Apple.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 6.16 has nine time lines on a daily Apple chart, each rising from a circle where volume tests a lower volume threshold. Follow each line up to price: some are tops and some are bottoms.

2.5 Overbought or oversold only afterwards

Extreme buying and extreme selling both generate volume, so an overextension in volume can signal either a top or a bottom. That is why the book will not call a volume extreme overbought or oversold until it can be tied to a top or a bottom. We can pinpoint the overextension when it happens, but the overbought or oversold reading comes only in retrospect. Overbought and oversold were met in Chapters 1 and 5 and are used here without a definition; oscillators are the book's Chapter 8.

In Chart P the two volume records are identical. Only what price did tells them apart: after a rise the spike was a top, overbought in retrospect; after a fall, a bottom, oversold in retrospect.



Not prices: sketches. The two volume records are identical; only price tells them apart.

Chart P The same volume extreme after a rise and after a fall.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

2.6 The timeframe changes what volume shows

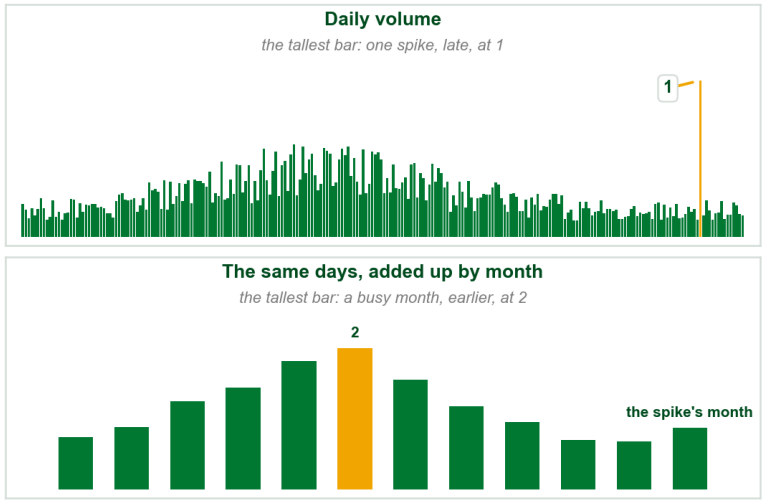
The continuous chart of gold with five volume panels under it: daily, weekly, monthly, quarterly and yearly. Lines over each mark where its volume peaks: in 2013 on the daily and weekly panels, in 2011 on the monthly, quarterly and yearly ones.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 6.17 Gold's volume on five timeframes.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 6.17 shows the same gold volume on daily, weekly, monthly, quarterly and yearly panels. Daily and weekly volume peak in 2013; monthly and higher peak in 2011. **Lower timeframe charts are more sensitive to volume action, and at higher timeframes the overall trend or behavior of volume is more obvious.**



Invented volume: 240 trading days, then twelve months of twenty days each.

Chart Q One volume record by day and by month.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart Q shows why. Its upper panel is 240 invented days, whose tallest bar is a single spike late in the record, at 1. Its lower panel is the same days added up month by month, and the busiest month, at 2, comes earlier: one day's spike is outweighed by the busy months around it.

CHECK YOURSELF

A volume spike towers over everything around it. Why can you not call it overbought on the day it happens?

Section 3

Price Barriers and Filtering

What volume says about the strength of a support or a resistance, the five kinds of volume action at a barrier, the four ways to filter volume, and trendlines and channels drawn only from heavily traded points. The book's section 6.1, pages 187 to 190.

3.1 Volume and the strength of a barrier

Chapter 5 called a support below price or a resistance above it a barrier. The book now asks which barriers to trust. **Support and resistance formed at a volume peak are regarded as the most reliable and the strongest barrier to future retests.** The reason it gives: even with the surge in volume, price could not get through, which gives traders confidence that the barrier will hold, and since expectation is everything in the market, that confidence goes a long way. A barrier formed across a volume dip or trough is less reliable: it held only because nobody was interested in taking price through it.

Figure 6.18 poses the question in four sketches. Resistance 1 and Support 1 are touched as volume peaks and are the more reliable; Resistance 2 and Support 2 are touched as volume dips.



Figure 6.18 Which price barrier is more reliable?

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

The five kinds of volume action at a barrier

1. Volume increasing across support or resistance.
2. Volume decreasing across support or resistance.
3. Volume peaking at support or resistance.
4. Volume bottoming at support or resistance.
5. Volume flatlining across support or resistance.

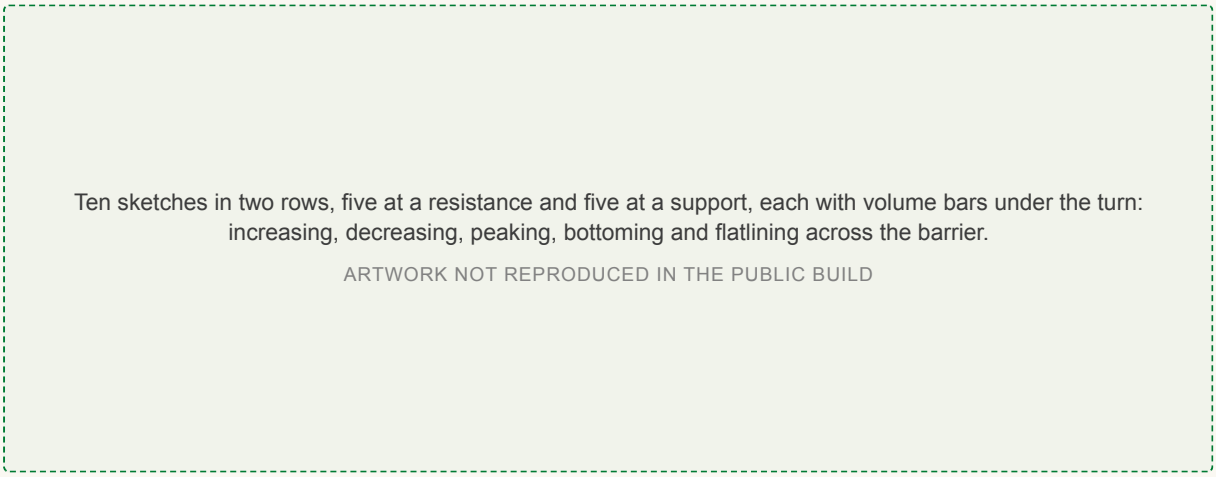
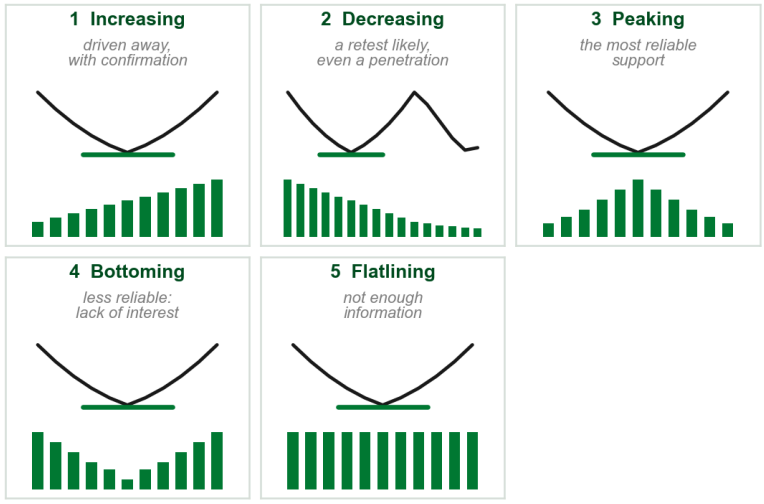


Figure 6.19 The five kinds at a resistance and at a support.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 6.19 draws all five at a resistance, top row, and at a support, bottom row. Increasing or decreasing volume across a barrier signifies a change in sentiment there. With increasing volume, volume picks up as price reverses at the barrier and drives price away from it with bullish or bearish confirmation. With decreasing volume, there seems too little force to drive price through, and volume falls further as price reverses. **Decreasing volume across a barrier tends to see a retest, and in more extreme circumstances price may return to penetrate it.** Peaking is the most reliable case and bottoming the less reliable, as above, and flatlining volume does not give enough information.

Chart R sketches the five at a support, with the book's outcome written under each name. Read each case as what volume does, then what price does next: in the decreasing case, price drifts back down for a retest.



Not prices: sketches. Each price falls to a support, the green line, and turns up.

Chart R What each of the five tells, at a support.

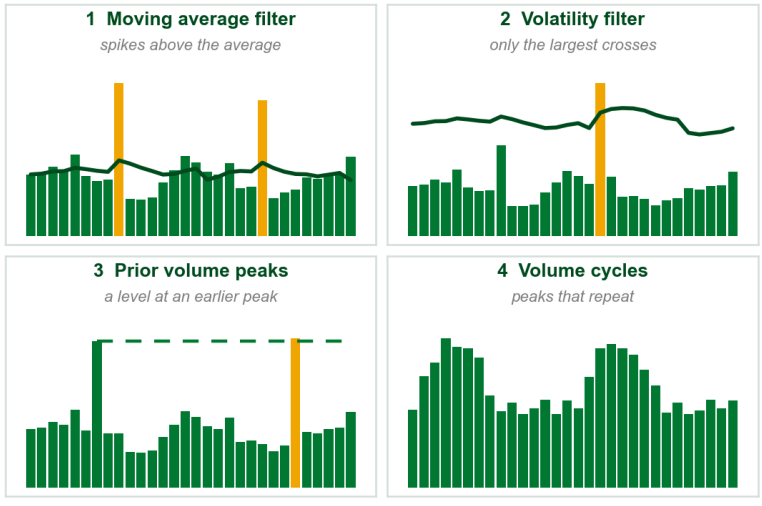
FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

CHECK YOURSELF

Two supports have held. One held on the heaviest volume of the month, the other on the lightest. Which would the book trust more on a retest, and why?

3.2 Four ways to filter volume

Filtering picks out the most significant volume action. The book lists four ways: a moving average filter, a volatility filter, prior volume peaks and volume cycles. Prior volume peaks are the overextended levels of Section 2, and volume cycles come in Section 5. This is review question 5.



Invented prices and volume. Gold: the volume each filter picks out.

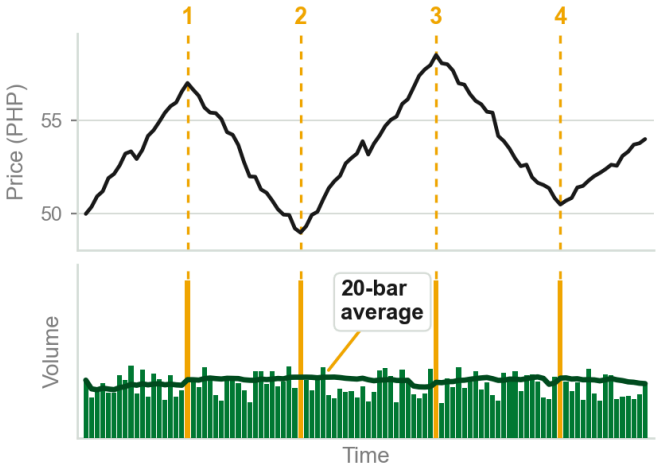
Chart S One sketch of each of the four filters.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart S sketches each filter on invented volume, with the volume each picks out in gold: spikes above a moving average, the largest bar above a higher band, a bar that reaches the level of an earlier peak, and peaks that repeat in a cycle.

The first is a moving average of volume, a line of the average of recent volume drawn over the volume bars, so that a spike well above it stands out.

> **Smoothed volume filter.** A smoothed volume filter is simply a moving average based on volume, instead of price. Volume spiking above this overextended moving average line is considered significant, and may lead to a top or bottom.



Invented. Line: the average of the last 20 volume bars, computed. Gold: four spikes.

Chart T A 20-bar average of volume, and four spikes far above it.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

In Chart T each point of the line is the average of the last 20 volume bars, computed. Four spikes stand far above it, in gold, and the dashed lines numbered 1 to 4 run from each up to a top or a bottom in price. The book gives no measure of how far above the line a spike must be.

How long should the average be? The book says a simple backtest may be needed to find the lookback period right for a setup, and that some practitioners set it to the dominant trading cycle period, or half of it. A backtest, as Chapter 5 glossed it, runs a method against past prices, and the cycle period is the Chapter 5 term for how long each swing takes. The book gives no number.

The daily USDJPY chart with a smoothed volume line across its volume panel, labelled moving average overextension threshold. Dotted arrows rise from the overextensions in volume to tops and bottoms in price.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

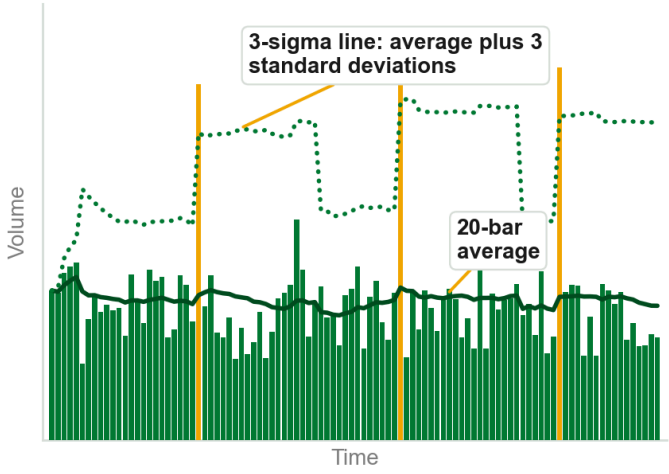
Figure 6.20 A smoothed volume filter on USDJPY.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 6.20 draws the smoothed volume line, labelled moving average overextension threshold, across the volume panel of daily USDJPY. Follow the dotted arrows from the overextensions in volume up to tops and bottoms in price.

The second is the volatility filter. Sigma is the Greek letter for a standard deviation, which Chapter 5 glossed as how far values typically spread from their average. A line two or three standard deviations above average volume lets only the biggest surges through. The book builds it from a Bollinger Band, which Chapter 5 named and this chapter does not explain; only the band's upper line is needed.

Two- or three-sigma volume filter. A two- or three-standard deviation, or sigma, filter, also called a volatility filter, keeps only the most significant surges: a 2 or 3 standard deviation Bollinger Band on volume, upper band only.



Invented volume. Both lines are computed over the last 20 bars. Gold: the bars above the 3-sigma line.

Chart U A 20-bar average of volume and a three-sigma line above it.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart U draws both lines on invented volume: the 20-bar average, and above it the three-sigma line, the average plus three standard deviations of the same 20 bars. Only three bars cross the upper line, in gold. A three-sigma line is stricter than a two-sigma one, so fewer bars cross it.

The daily USDJPY chart with a stepped line across its volume panel labelled 3-sigma volume overextension threshold, and dotted arrows from the few overextensions in volume above it to tops and bottoms in price.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 6.21 A three-sigma volume filter on USDJPY.

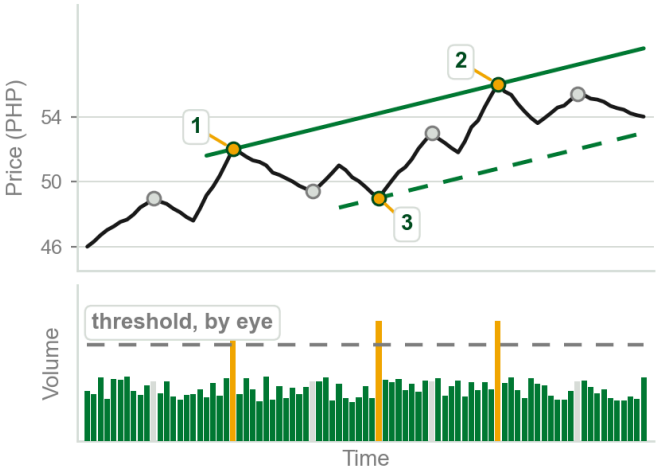
Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 6.21 is the same USDJPY chart with a three-sigma threshold: there are fewer arrows than in Figure 6.20, at the most significant tops and bottoms. The book notes that a standard deviation oscillator could be used instead, but it is less useful, because it gives no overlay on which volume can be seen crossing the threshold.

3.3 Volume filtered inflection points

Chapter 5 taught inflection points, a peak or a trough, and drew trendlines and channels through them. Many practitioners use volume to choose which inflection points to draw through: only those that came with heavy volume.

> **Volume filtered inflection points.** Many practitioners use only peaks and troughs that are considered significant, that is, associated with above average volume, to construct trendlines and channels.



Invented prices and volume. Solid: the line through 1 and 2. Dashed: its parallel through 3. Grey: turns on too little volume.

Chart V A channel drawn only through heavily traded turns.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

In Chart V six turns came on light volume and are greyed out. Three came on volume above the threshold and are numbered 1, 2 and 3. The solid line joins 1 and 2, and the dashed line is its parallel through 3: a channel built only from significant points.

The daily silver chart, XAGUSD, with a tick volume overextension threshold across its volume panel. Arrows rise from the circled tick volume peaks to points 1, 2 and 3; a channel is drawn from 1 and 2 with a parallel from 3, and price reacts at X.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 6.22 A channel from tick volume filtered points in silver.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 6.22 uses tick volume, a count of trades defined in Section 4, with a threshold drawn by visual inspection so that only the most significant peaks reach it. Points 1, 2 and 3 came with overextended tick volume. A trendline is drawn from 1 to 2 and a parallel from 3, and at X price reacts to the channel, a short-term buying opportunity, before the market turned down through it.

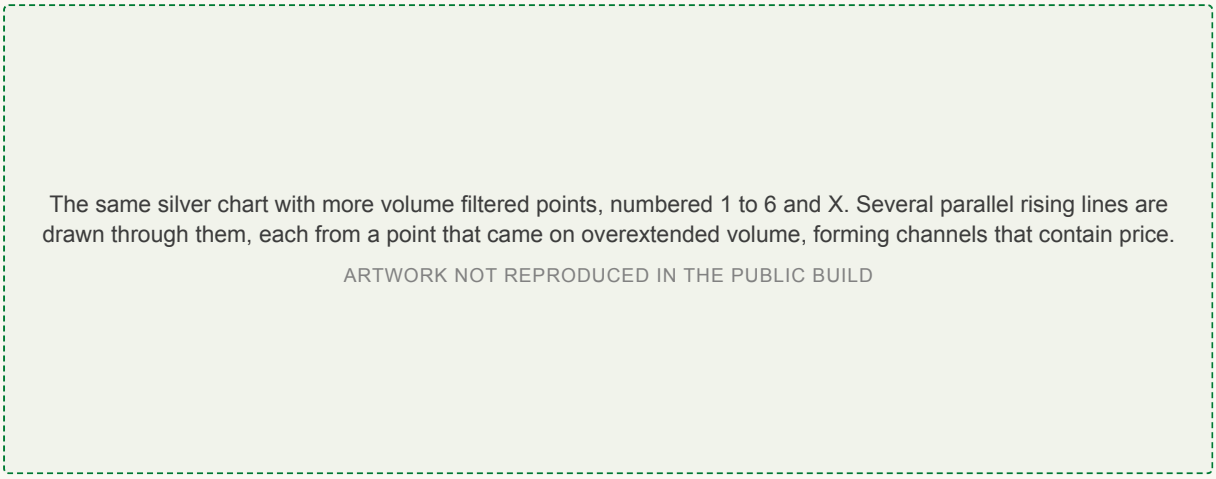


Figure 6.23 Further channels from filtered points in silver.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 6.23 builds more channels on the same chart. A parallel from 3 is breached by a large bearish candlestick; a parallel from 4, taken back in time, meets the trough at X, which the book admits was found in retrospect but still calls a valid rising channel; parallels from 5 and 6 contain price as resistance and support. The book calls it near perfect angular symmetry, a Chapter 5 term. Its text calls Figure 6.23 the same daily chart of USDJPY and then the Silver market; both captions and charts say silver.

Section 4

Relative Volume, Open Interest and Breadth

Relative against absolute volume, why bull markets trade more, open interest and how it is read, tick volume, the TRIN, and volume by price against volume by time. The book's section 6.1, pages 190 to 197.

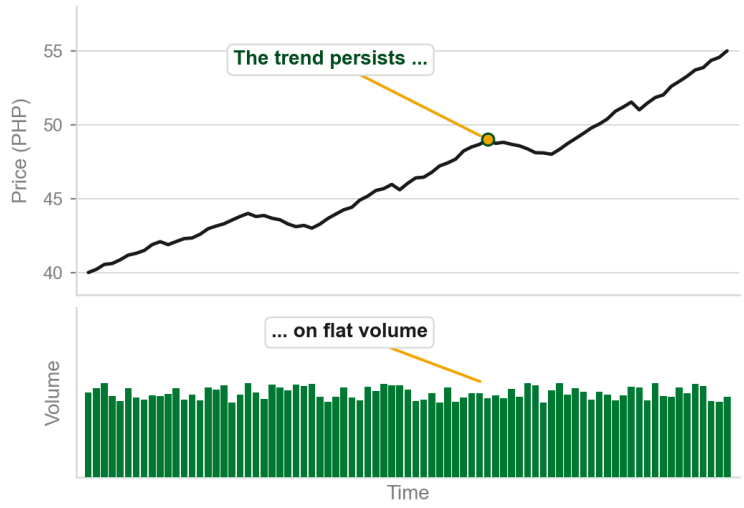
4.1 Relative and absolute volume

Absolute volume is not as important as relative volume: it is the change in volume that gives the most information. Rising and declining volume show the interest at various prices and whether a trend is bullish or bearish, and historical peaks and troughs in volume, the overextensions, give a reference level for future tops and bottoms. This is review question 7.

The book adds a caution: a change in volume is not an absolute requirement for a trend to develop. A trend can persist for long periods on flat volume.

Chart W is that caution drawn: an invented share rising for the whole chart while every volume bar stays about the same height.

Relative volume also shows a difference between bull and bear markets, and the first of the book's reasons for it is a name of its own. Bought at 50 pesos and now at 38, a holder may keep the share, waiting for 50 again, rather than sell at a loss.

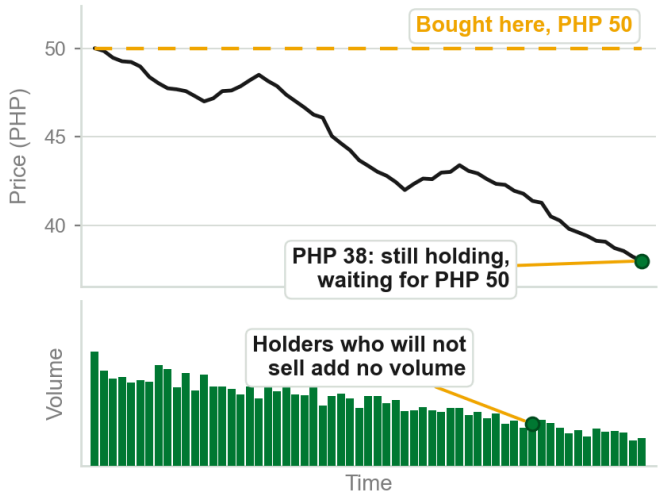


Invented prices and volume. Volume stays about the same all the way.

Chart W A trend that persists on flat volume.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Loss aversion bias. Many investors hold on to losing positions in a falling market, hoping they rebound to breakeven or profitability, believing unrealized losses are not real losses: loss aversion bias.



Invented prices and volume. One holder's story; the pesos are ours.

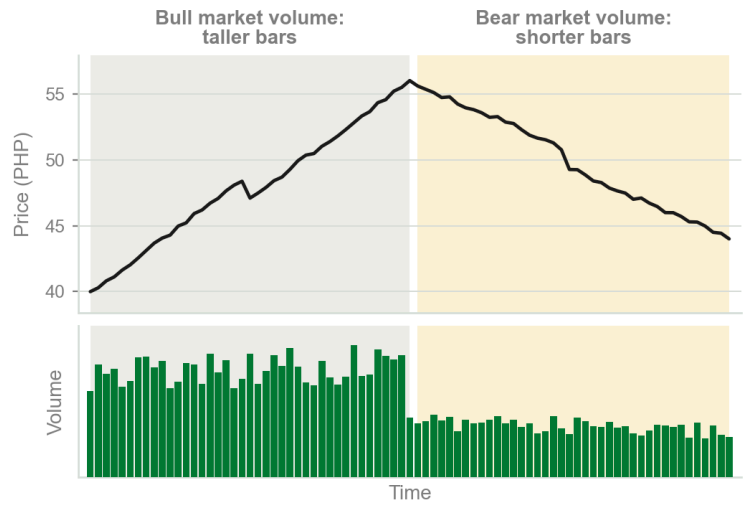
Chart X A holder waiting for breakeven while volume shrinks.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

In Chart X the dashed gold line is the 50 peso entry. Follow the price down to 38 and the volume under the fall shrinking: holders who will not sell add no volume. Chapter 1 listed loss aversion among the behavioral biases in one line; this is where it is explained.

Why bull market volume outweighs bear market volume

- Loss aversion: holding on to losing positions in a falling market does not contribute to its trading.
- In a rising market, investors and traders can use house money, their profit, to fund new positions.
- More investment funds market their services in a bull market, drawing more investors in.



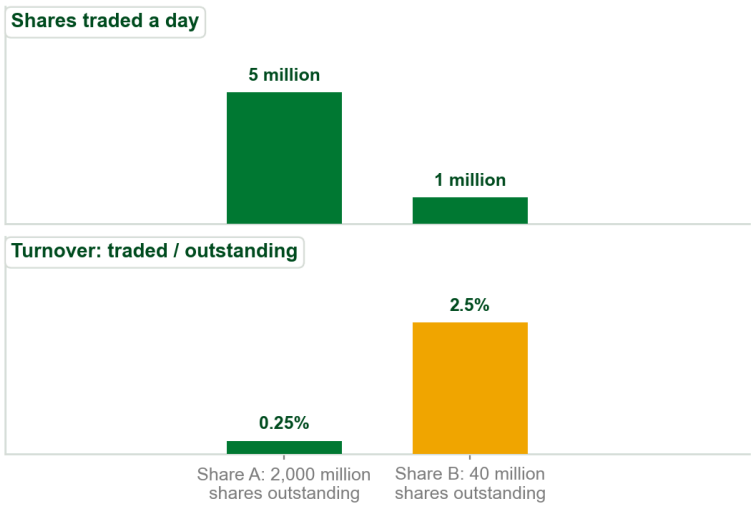
Invented prices and volume. The book gives three reasons; the chart shows only the result.

Chart Y Bull market volume against bear market volume.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart Y shows only the result: the volume bars under the rise are taller than those under the fall. The book says these are a few of a variety of reasons.

Absolute volume has its own uses. It helps find potential price rejection levels, a Chapter 3 term, at significant or historical prices. It compares participation across stocks, sectors, industries or commodities in search of greater liquidity, the book's word, not defined here. And total turnover against the shares outstanding gives the absolute level of participation in a stock.



Our example: two invented shares.

Chart Z Turnover: two invented shares compared.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

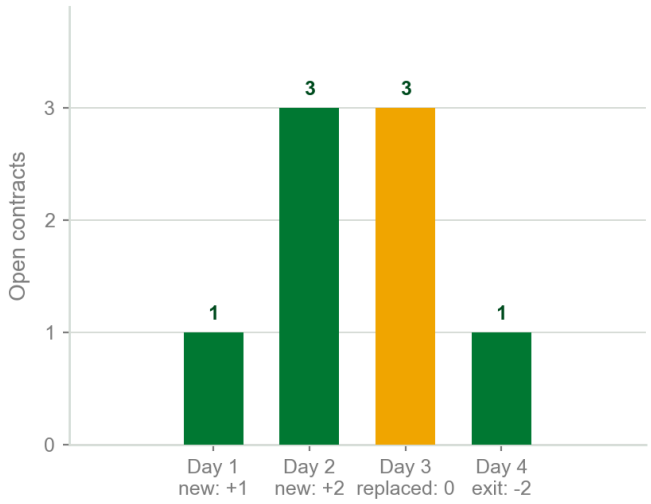
Chart Z works turnover for two invented shares. Share A trades 5 million a day of 2,000 million outstanding, 0.25 percent. Share B trades 1 million of 40 million, 2.5 percent. A trades more shares; B has ten times the participation.

Finally, volume distortions. A surge in arbitrage, program trading and high frequency trading may distort the traditional meaning of volume, so the volume traders read may not be totally indicative of true volume action in the classical sense. Chapter 1 named all three; the book explains none of them here and gives no remedy.

4.2 Open interest

Chapter 3 listed open interest as transaction data and Chapter 4 used it in a consolidation; this is where it is defined. Unlike stocks, all futures and options contracts eventually expire, and each contract is one long position and one short position. A contract stays open until both sides are out. Suppose A buys one contract from B, both new: open interest is 1. C buys two from D, both new: 3. A sells his to a new buyer, E: still 3, because one side was replaced. D buys two back from C, both leaving: 1.

> **Open interest.** The total amount of outstanding contracts in the futures and options markets. It increases when a new long and short are initiated, decreases when both exit, and is unchanged when one is replaced.



Our example: traders A to E, one contract market.

Chart AA Open interest over four days.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart AA draws those four days as columns of 1, 3, 3 and 1 contracts, each labelled with what moved it. Only both sides arriving together, or both leaving together, changes the count.

Generally, open interest is interpreted the same way as volume: it shows the level of interest in seeing price rise or fall, and its divergence is read the same way as for non-price-based volume oscillators.

Price, volume and open interest together

- Price, volume and open interest all rising: bullish, predominantly driven by buying pressure.
- Price rising while volume and open interest decline: bearish, driven by shorts covering and longs leaving.
- Price falling while volume and open interest rise: bearish, driven by short selling.
- All three falling: bullish, as fewer longs are liquidated and fewer shorts initiated; participants are less interested in seeing the market decline further.

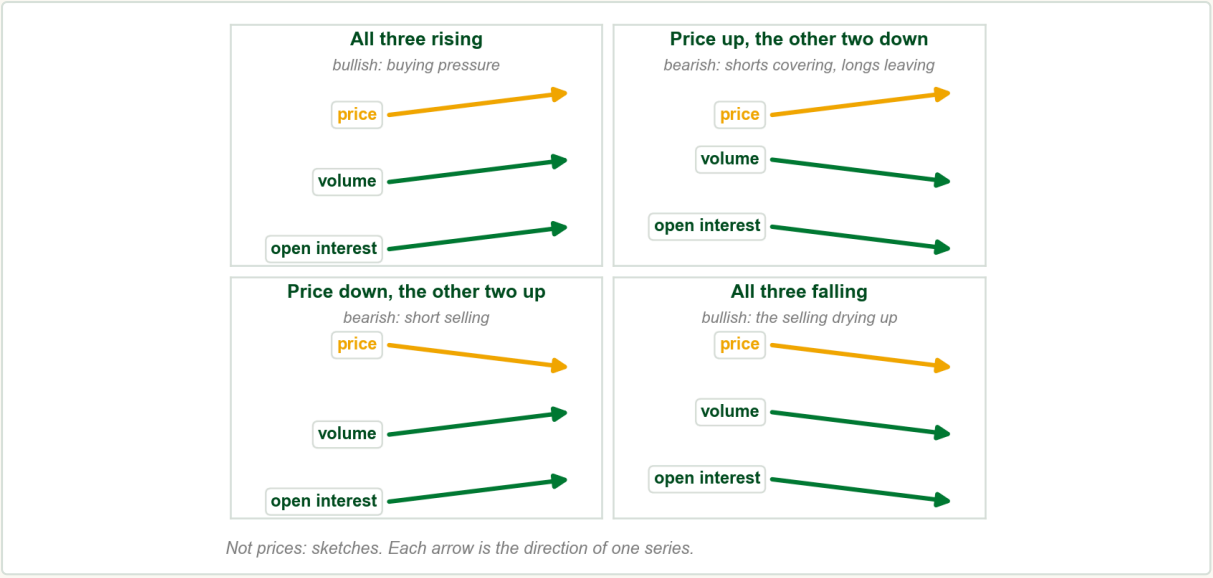


Chart AB The four cases as arrows.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart AB draws each case as three arrows, price in gold. The rule is Section 1's with open interest alongside volume: with price, bullish; against it, bearish. This is review question 6.

The trader's actions follow from the four. With price, open interest and volume all rising, look for buy signals. With rising prices on declining open interest and volume, look to exit and liquidate longs. In a declining market with open interest and volume rising, look to go short, and rapidly cover all shorts when both start to decline together. Going long, liquidating, going short and covering are Chapter 1 terms.

The continuous daily chart of silver rising from A to B, with an open interest panel showing a lower peak at 2, labelled bearish divergence, and a volume panel showing a higher peak at 1, labelled bullish confirmation.

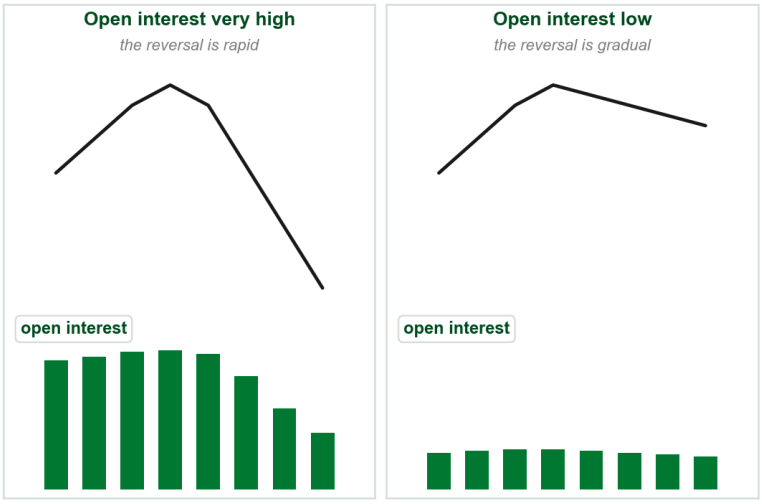
ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 6.24 Open interest diverging from silver's rise.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 6.24 is silver rising from A to B. Volume peaks with the price at 1, bullish confirmation, but open interest makes a lower peak at 2, bearish divergence: many longs are liquidating as prices rise. Silver then retraced significantly.

The level of open interest may also indicate the nature of a reversal or breakout. If it is very high, the reversal may be significantly more rapid, because of the large number of open contracts liquidating and covering.



Not prices: sketches. The open interest panels share one scale. The book says may: a tendency.

Chart AC High open interest and a rapid reversal, low and a gradual one.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart AC sketches the two on one scale: the fall after a top on very high open interest is steep, and on low open interest gradual. May, says the book: a tendency.

CHECK YOURSELF

A trader holding a long futures contract sells it to a newcomer who takes over the long. What happens to open interest, and why?

4.3 Tick volume

Chapter 1 met tick volume in a caption, as transactions counted regardless of size, and Chapter 3's tick chart drew a new bar after a set number of trades. The book now defines it. One tick is one trade, and one trade may be 1 contract or 1,000. In the over the counter, OTC, foreign exchange market many brokers offer tick volume instead of actual volume. The five trades of Section 1.3 give a volume of 2.4 million shares and a tick volume of 5. This is review question 1.

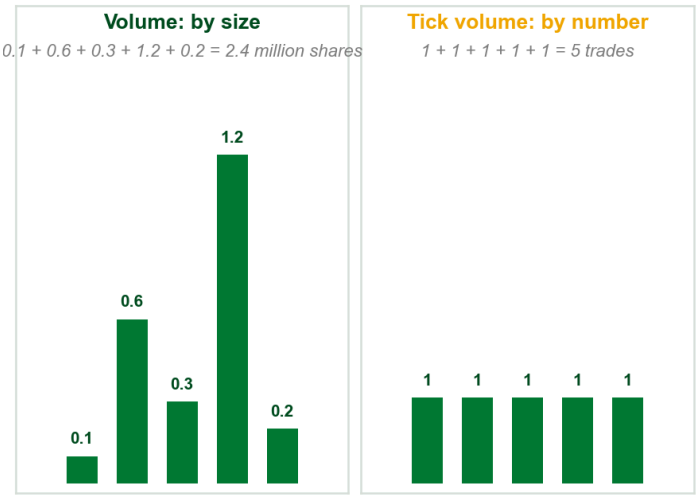
> **Tick volume.** One tick represents one trade: tick volume is transaction volume. Many OTC fx brokers offer it instead of actual volume; strongly correlated with it, it may be used as a proxy.

Chart AD counts the five trades both ways: by size, columns of 0.1 to 1.2 million adding to 2.4 million shares; by number, five equal ticks adding to 5. The book says tick volume may serve as a proxy for actual volume as far as comparing relative levels of overbought and oversold goes.

4.4 Breadth volume and the TRIN

The Arms Index, or TRIN, sets breadth against volume. Advances and declines were listed as breadth data in Chapter 3 and never defined; read them here as the number of stocks rising and falling, and up volume and down volume as the volume in each. That reading of the names is ours. With 1,500 advances and 1,000 declines the first ratio is 1.5; with up volume of 600 million and down volume of 800 million the second is 0.75; the TRIN is 1.5 divided by 0.75, which is 2.0.

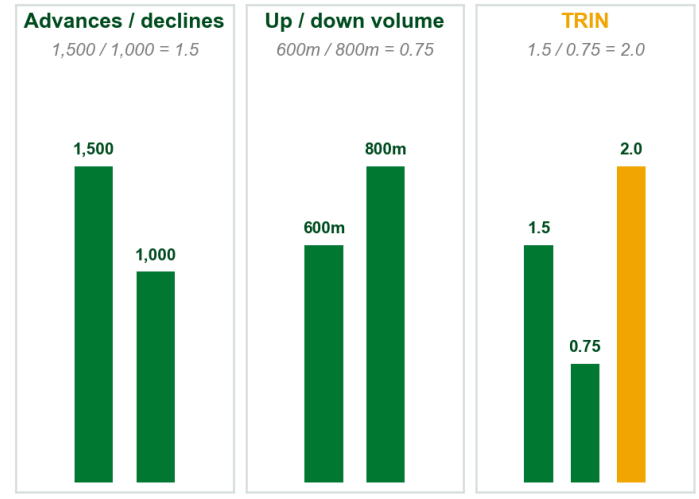
> **The Arms Index, or TRIN.** The Arms Index, or TRIN, incorporates breadth volume, advances and declines: TRIN = (advances / declines) / (up volume / down volume).



Our example: Part 1's five trades, counted two ways.

Chart AD The same five trades counted by size and by number.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.



Our example: invented counts and volumes.

Chart AE The TRIN worked as arithmetic.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart AE sets the sum out in three columns: advances over declines, up volume over down volume, and the first divided by the second.

The book reads the TRIN together with a second indicator it calls the tick, and never says what that tick measures; it is not shown to be the tick volume above. Its short-term rule: a rising but not overbought tick with a falling but not oversold TRIN is bullish, and the converse bearish. Its longer-term rule: a tick that is oversold, above its two-sigma level on a weekly chart, with a TRIN that is overbought, near or above 3.0 on the daily chart, is bullish, and the converse bearish.

Figure 6.25 is a 1-minute chart of the S&P 500 index, though its caption says daily. The box marks the tick and the TRIN both at oversold levels, and the index rose. **Here the book says two things: its text calls a bullish TRIN overbought, and its figure calls its bullish TRIN reading oversold.** Both readings are bullish; the word differs. Overbought and oversold are used without a definition, as in Section 2.

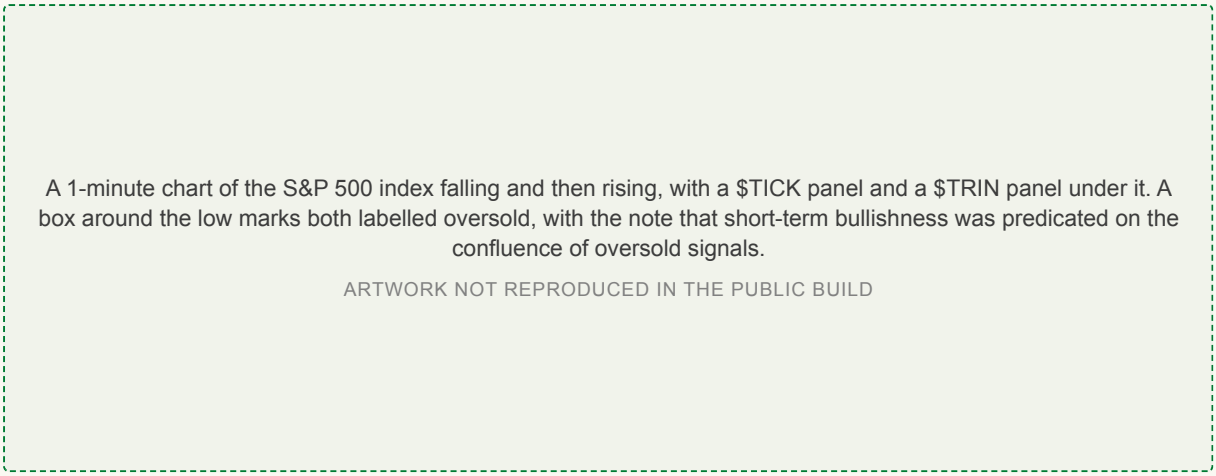


Figure 6.25 The tick and the TRIN both oversold on the S&P 500.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)



Figure 6.26 A bearish rising TRIN on the S&P 500.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

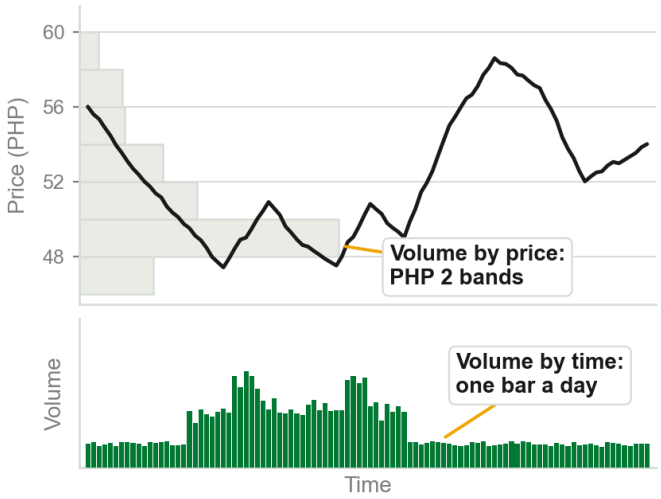
Figure 6.26 shows a short-term downtrend in the S&P 500 with a bearish rising TRIN, boxed, and a tick largely absent of activity between 1:45 p.m. and 2:45 p.m. Together, the book calls it a short-term bearish signal. The figure also labels balanced tick action as a potential consolidation, which the text does not discuss.

4.5 Volume by price and volume by time

Volume may be analyzed two dimensionally. The usual volume bars, one a period, are volume by time. Volume by price adds up all the volume traded at each price level and draws it as bars running sideways from the price axis. If a share trades 9 million shares between 48 and 50 pesos in three months and 2 million between 54 and 56, its longest sideways bar is at 48 to 50.

> **Volume by price and volume by time.** Volume by time represents the total volume traded over a specified time interval or period. Volume by price identifies the total volume traded at each price level.

Chart AF draws both on one invented share. Each sideways bar is the volume of every day that closed in its two-peso band; the bars under the chart are one a day.



Invented. Each sideways bar: the volume of the days that closed in its band.

Chart AF Volume by price and volume by time on one chart.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chapter 4's Figure 4.19 showed the same thing and called it price-based volume. That is this chapter's volume by price, and not the price-based volume oscillators of Section 1, which build price into a line.

The daily iShares Silver Trust with volume by price drawn as horizontal bars from the left edge. The longest bars sit around the $17 level, labelled significant capital commitment, strong support, and another cluster at the newer consolidation level higher up.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 6.27 Volume by price marking support in silver.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 6.27 shows a large volume turnover around the $17 level in the iShares Silver Trust: find the longest sideways bars and then price returning to that level. That $17 is a significant support, and this is not obvious from conventional volume bar action.

The book's four steps to support and resistance

1. Locate the largest volume by price range.
2. Locate the largest volume by time intervals.
3. Find the area where the two intersect.
4. The closest peak or peaks above that area is potentially the most significant resistance; the closest trough or troughs below it, the most significant support.

Figure 6.28 brackets a lot of volume turnover between $77 and $86 on the daily chart of 3M, the most traded range, with some resistance at $90 to $92.

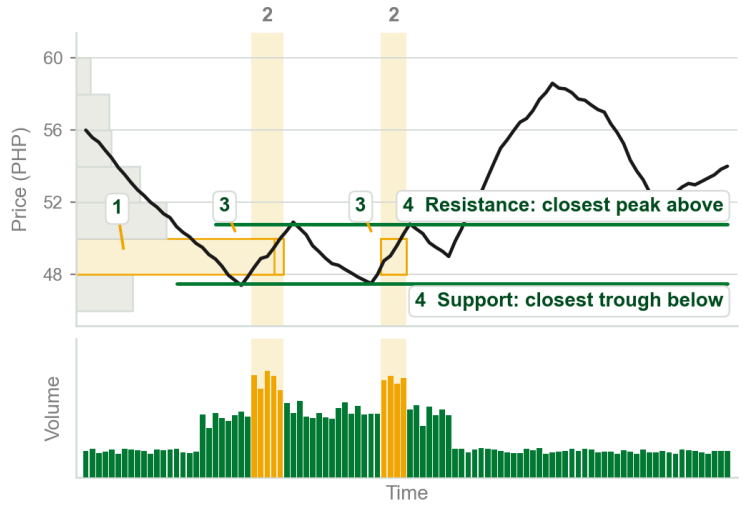
The daily chart of 3M with volume by price as horizontal bars, the longest bracketed between about $77 and $86 and labelled maximum capital commitment in this range, an important support zone in the future.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 6.28 The most traded range on 3M.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

These price ranges are potential future support and resistance levels, and the four steps above pin the most significant of them down.



Invented. 1, 2 and 3 are the book's first three steps; 4 is the two levels.

Chart AG The four steps, numbered on one invented chart.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart AG numbers the steps on the chart in the book's order. Step 1 is the gold band, the largest volume by price. Step 2 is the two shaded stretches, the busiest days. Step 3 is the two small gold boxes where they cross. Step 4 is the two lines: resistance at the nearest peak above the boxes and support at the nearest trough below.



Figure 6.29 The four steps on 3M.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 6.29 does the same on 3M: the $74 to $80 band is the largest volume by price, the dark columns are the largest volume by time, the two white boxes are where they cross, and the dashed arrows mark the closest resistance above and support below. Both acted as effective barriers to price.

CHECK YOURSELF

Advances 1,200, declines 800, up volume 900 million, down volume 600 million. Work out the TRIN.

Section 5

Bar Range, Volume Cycles and Non-Constant Time

The four named bars of bar range against volume, the constant volume chart, whether volume leads price, the intraday volume cycle, and what a non-linear time axis does to a line. The book's section 6.1, pages 197 to 202.

5.1 Volume against bar range

Chapter 5's characteristic 15, volume spread action, set a bar's range against its volume four ways. The book's heading here says spread too, meaning bar range as in Chapter 5, and not Chapter 3's bid-ask spread. In this chapter the four cases forecast potential breakouts and reversals, and each bar gets a name.

Figure 6.30 is the book's whole picture of them: for each of four numbered cases, an empty box for the bar range above a black box for the volume. Narrow and wide, low and high, are against the trend's usual bars and volume; the book gives no number for either. Suppose a fast uptrend's usual bar is one peso on 2 million shares: a 20 centavo bar on 5 million shares is a narrow range on high volume.

> **Rest, reversal and continuation bars.** Narrow range on low volume: Low Volume Rest Bar. Narrow, high volume: High Volume Reversal Bar. Wide, low volume: Low Volume Reversal Bar. Wide, high volume: High Volume Continuation Bar.

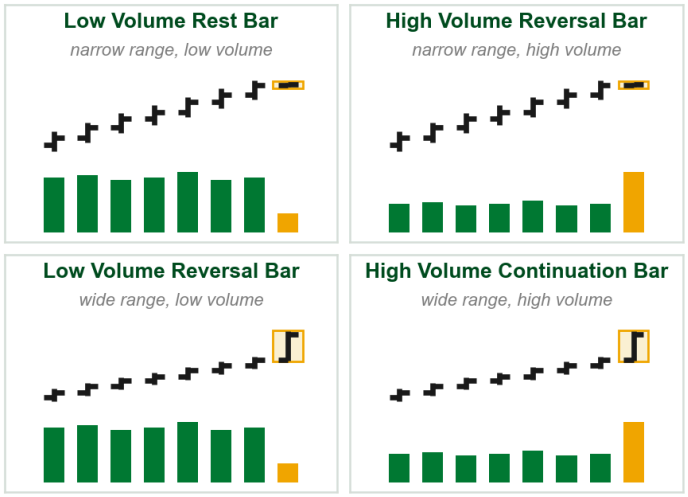
Chart AH ends each of four short uptrends in one of the named bars, boxed in gold, with its volume in gold beneath.

Four numbered pairs, each an empty box for the bar range above a black box for the volume: 1 a short range on low volume, 2 a short range on high volume, 3 a tall range on low volume, 4 a tall range on high volume.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 6.30 The four combinations of bar range and volume.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)



Not prices: sketches. Seven ordinary bars, then the named one, boxed in gold.

Chart AH The four named bars, each at the end of a short uptrend.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

What each bar indicates

- Low Volume Rest Bar: a lack of participation. In a strong and rapid trend, a potential reversal; in a weak to average trend, a point of rest.
- High Volume Reversal Bar: a strong potential for reversal. If the volume represented buying in an uptrend the range would be much wider, so sellers are attempting to take control. Bearish in an uptrend; the converse in a downtrend.
- Low Volume Reversal Bar: a large move on relatively low participation, a lack of interest. The longer the bar, the more bearish in an uptrend and bullish in a downtrend.
- High Volume Continuation Bar: a large move backed by very large participation, interest in seeing price continue. Bullish in an uptrend, bearish in a downtrend.

Chart AI draws the two reversal bars: on the left a narrow bar on very heavy volume at the top, on the right a very long bar on light volume, and price falls after each. One is too little movement for the volume, the other too little volume for the movement. Chapter 5 called a large range on low volume only trend inhibiting; Chapter 6 calls it a reversal bar.

Chart AJ draws the same rest bar in two trends.

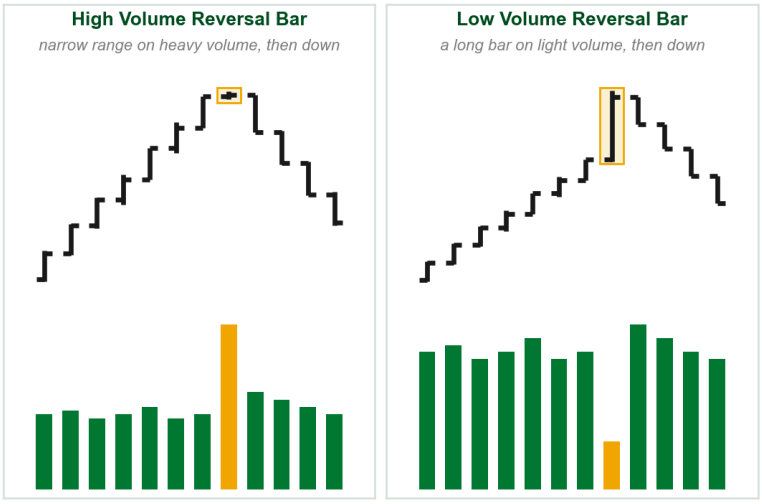


Chart AI The two reversal bars.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

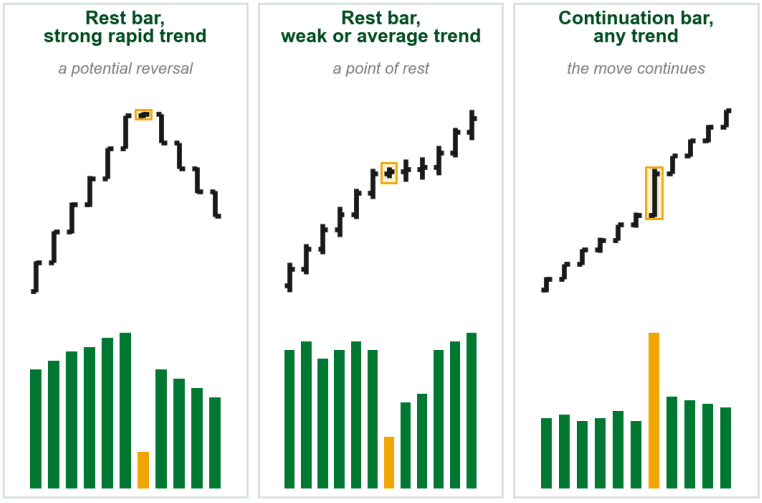


Chart AJ The rest bar in two trends, and the continuation bar.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

At the top of a strong, rapid rise price reverses after it; in a slow, average rise price rests and rises again. The third cell is the continuation bar, wide on high volume, after which the rise continues. **The rest bar depends on the trend it is in; the continuation bar does not.**

Figure 6.31 is the PowerShares DB U.S. Dollar Bullish Fund with two narrow range candlesticks on high volume at Points 1 and 2; price reverses rapidly after each. On the left it also shows a low volume breakout signal near the apex of a descending triangle in August, the point where the triangle's two lines meet, and a large volume surge on the breakout.

Figure 6.32 is the converse case: a narrow range candlestick on high volume at a low around the beginning of November, followed by a significant rebound.

The daily PowerShares DB U.S. Dollar Bullish Fund. Narrow range candlesticks on high volume are circled at 1, near a low, and at 2, near a high, with lines down to their tall volume bars. At the left a triangle narrows into a low volume breakout signal.

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Figure 6.31 High volume reversal bars on a dollar fund.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

The daily PowerShares DB U.S. Dollar Bullish Fund with a narrow range candlestick on high volume circled at a low at the start of November, followed by a strong rise.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 6.32 A high volume reversal bar at a bottom.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

The daily PowerShares DB U.S. Dollar Bullish Fund in a long decline under its moving averages, with a circled high volume reversal bar at a short rally's top in late May, a falling line over volume and the MACD leading up to it, and a high volume continuation bar marked X.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 6.33 A reversal bar within the larger downtrend.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 6.33 shows a high volume reversal bar around 23 May 2011, after bearish divergence in volume and on the MACD, an oscillator Chapter 4 met and this chapter does not teach. **The book regards it as potentially more reliable because it formed within a larger downtrend.** At Point X there is a high volume continuation bar, and continuation bars too are more reliable in the direction of the predominant trend.

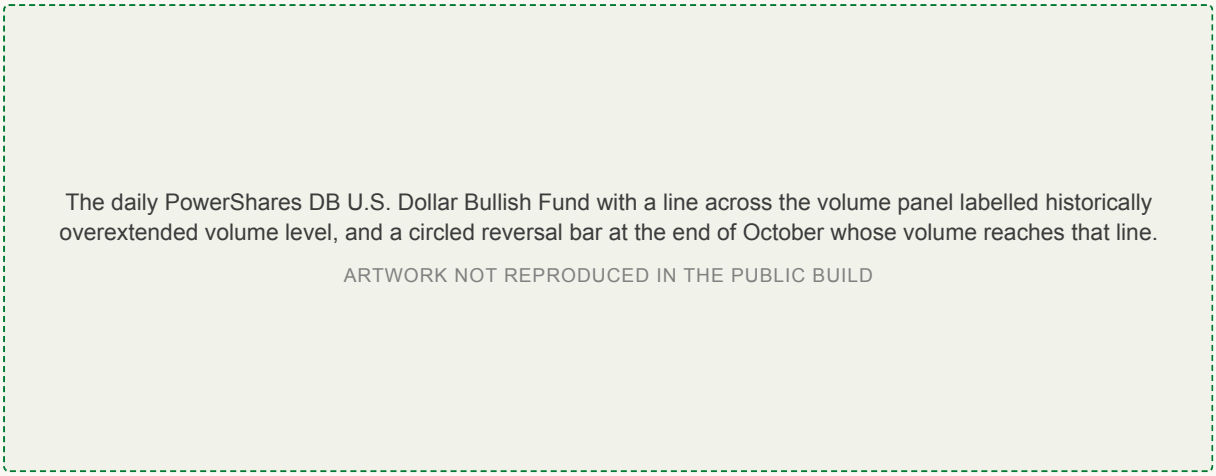


Figure 6.34 A reversal bar at a historically overextended volume level.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 6.34 shows a high volume reversal bar around the end of October 2011 whose volume reaches a historically overextended volume level drawn across the chart: two signals from this chapter agreeing at one bar.

5.2 Constant volume charts

Chapter 3 taught the constant volume chart, which draws a new bar each time a set volume has traded. Because every bar holds the same volume, a potential reversal bar can be found from the bar range alone and its height against the rest. A narrow bar there means a lot of volume for little movement; a very tall one, little volume for a lot.

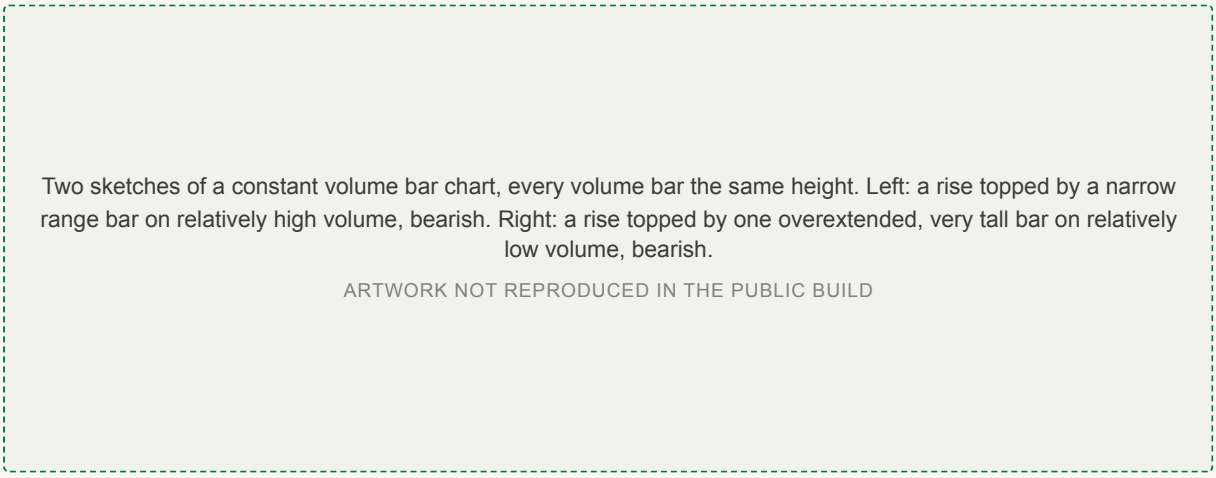


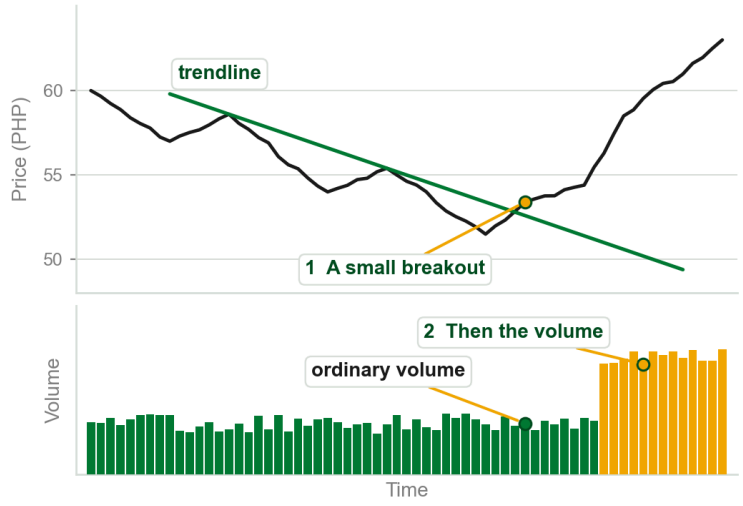
Figure 6.35 Reversal bars on a constant volume chart.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 6.35 sketches two rises on a constant volume chart, every volume bar the same height. On the left a narrow range bar on relatively high volume is bearish, sellers taking control; on the right a very tall bar on relatively low volume shows a lack of interest in the rise.

5.3 Does volume lead price?

Most practitioners agree that without participation, volume, price cannot move in a significant manner. Yet technical buy and sell signals need not have much volume to appear. A small breakout of a very significant trendline may set off a larger breakout, inviting tremendous volume as the penetration becomes obvious; then price leads volume. The book adds that with technical analysis so widely used, price may lead volume more often. Breakout here is Chapter 5's use of the word: price getting through a level, here a trendline.



Invented prices and volume. Price breaks the line first; the volume comes after.

Chart AK Price breaks the line first, the volume comes after.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

In Chart AK price breaks a falling trendline by a little at 1, on ordinary volume. Only at 2, as the breakout becomes obvious, does volume surge.

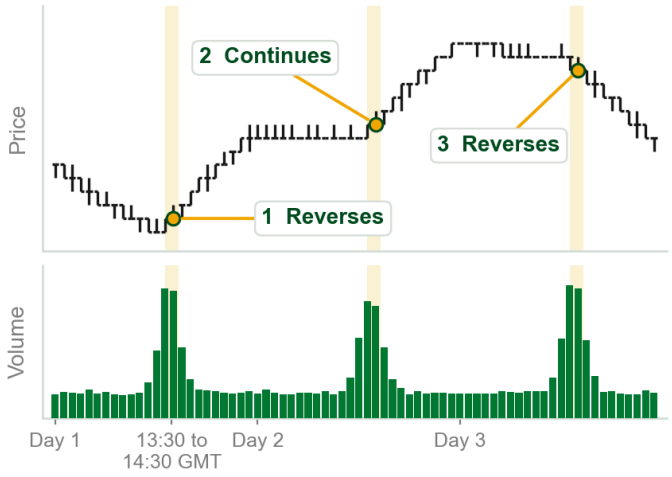
5.4 The intraday volume cycle

Volume also rises and falls on a daily rhythm, which the book uses to anticipate price. Its examples are EURUSD, and the peak it gives, 1:30 p.m. to 2:30 p.m. GMT, is 9:30 p.m. to 10:30 p.m. in Manila. The book does not say the hour holds for every market. This is review question 8.

> **Intraday volume cycle.** The intraday volume cycle normally peaks around 1:30 p.m. to 2:30 p.m. GMT. At these volume peaks, price will either reverse or continue, but very rarely remain unchanged.

Chart AL draws three invented days of hourly bars. Volume swells each day in the shaded peak hours; at the peaks, numbered 1 to 3, price reverses, continues and reverses again, and never stays where it was.

Figure 6.36 is a 30-minute EURUSD chart with its daily volume cycles along the bottom: almost every reversal comes at an intraday tick volume peak, which is especially useful when price retests a previous support or resistance. Many practitioners use a bidirectional, or straddle, entry at the peak, catching price whichever way it goes. Chapter 5 taught the bidirectional entry: a trade taken either way, depending on which way the breakout occurs.



Invented prices and volume. Hourly bars.
Gold: the peak hours, 13:30 to 14:30 GMT.

Chart AL Three days of hourly bars and their volume cycle.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

A 30-minute EURUSD chart over several weeks with support and resistance lines, a triangle and a channel, and daily volume cycles in the volume panel; price reverses at the volume peaks.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 6.36 Volume cycle peaks at support and resistance on EURUSD.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

An hourly EURUSD chart with three rising trendlines, retests marked at 1 to 5, a volume panel with daily volume cycles, and a stochastic panel whose oversold readings are marked under 1, 4 and 5, with circles at X and Y where it was not clearly oversold.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 6.37 Trendline retests at volume cycle peaks on hourly EURUSD.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

In Figure 6.37 price rebounds off trendline retests at Points 1, 4 and 5, each during a tick volume cycle peak, with a cycle-tuned stochastic oscillator showing oversold conditions.

At Points 2 and 3 the trendline failed, and there the stochastic showed no clear oversold condition, at X and Y. The cycle-tuned stochastic was named in Chapter 5; oscillators are the book's Chapter 8.

5.5 Volume on non-constant time charts

Point and figure, Renko and three line break charts, all Chapter 3 terms, do not plot the time axis in a linear fashion, and that changes the angle of trendlines, channels and chart patterns drawn on them. The equivolume chart, also Chapter 3, is unique: each bar is a constant time interval, but its width varies with volume, so its time axis is non-linear too.

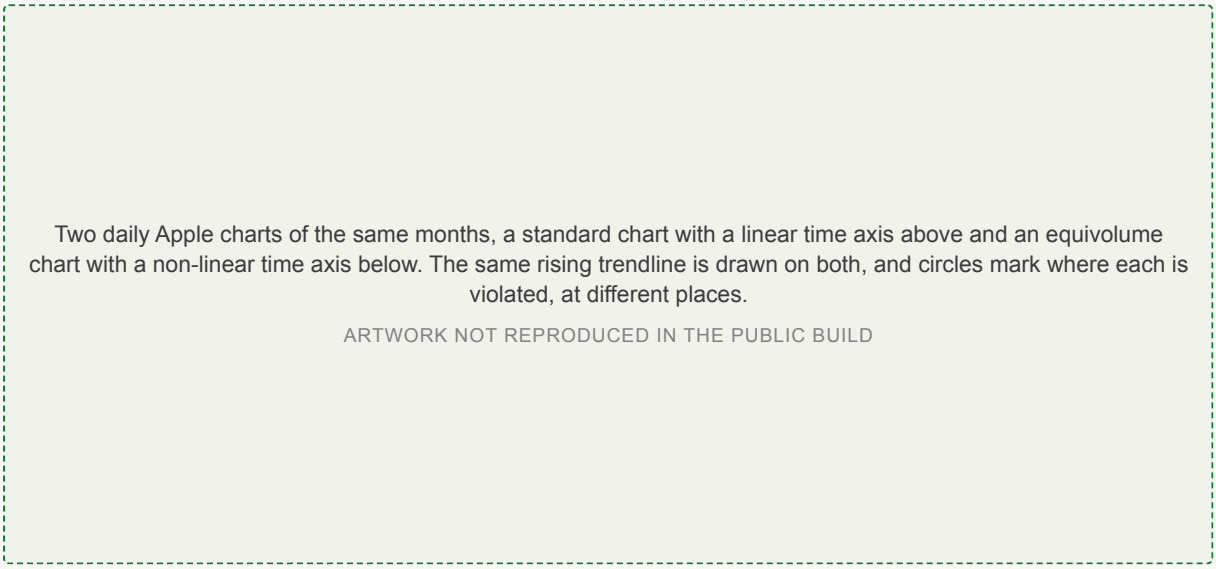


Figure 6.38 One trendline violated at different points on a standard and an equivolume chart.
Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 6.38 draws the same rising trendline on a standard daily Apple chart and on an equivolume chart of the same months. Compare the two circles: the line is violated at different places, because wide bars on heavy days stretch the equivolume chart's time axis.

CHECK YOURSELF

In an uptrend a bar prints a very narrow range on the heaviest volume of the month. Which named bar is it, and what does the book read from it?

Section 6

Volume Oscillators

The book's eight volume oscillators and the eight methods of reading them, the On Balance Volume and its weakness, the Accumulation Distribution Line, the two Chaikin indicators, the NYSE Up/Down Volume, the Herrick Payoff Index and the COT report. The book's section 6.2, pages 203 to 208.

6.1 Eight oscillators, eight methods

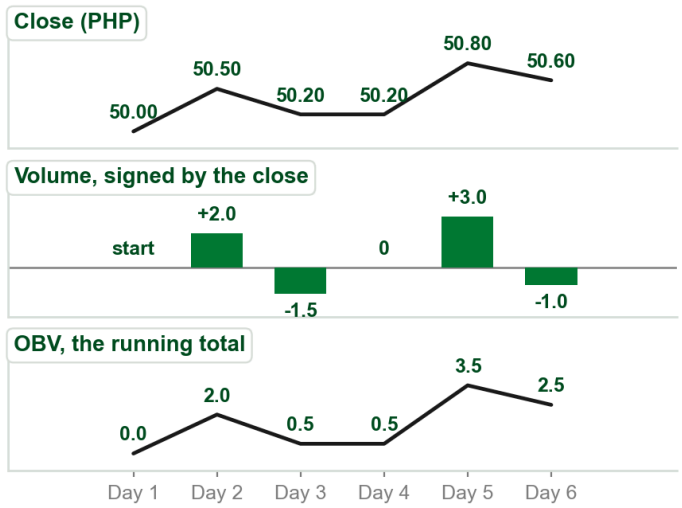
The book lists eight popular volume oscillators: the VWMA of Section 2, the Accumulation Distribution Line (ADL), On Balance Volume (OBV), the Chaikin Money Flow, the Chaikin Oscillator, the Demand Index, the Herrick Payoff Index and the NYSE Up/Down Volume Line. The Demand Index is never explained.

Each is read for bullish and bearish signals by eight standard methods of observation, which the book teaches in its Chapter 8: OBOS overextended levels, OBOS being its short form for overbought and oversold; equilibrium and zero line crossovers; signal line crossovers; price-oscillator divergence; geometrically, numerically and horizontally based oscillator pattern analysis; and oscillator-on-oscillator analysis. Only divergence is used in this chapter.

6.2 On Balance Volume

The OBV is a running total of volume in which only the direction of the close matters. Take closes of 50.00, 50.50, 50.20, 50.20, 50.80 and 50.60 pesos, with volumes after the first day of 2.0, 1.5, 1.0, 3.0 and 1.0 million shares. Day 2 closes up, so 2.0 is added; day 3 down, 1.5 subtracted; day 4 unchanged, nothing; day 5 up, 3.0 added; day 6 down, 1.0 subtracted. The OBV runs 0, 2.0, 0.5, 0.5, 3.5 and 2.5.

> **On Balance Volume, OBV.** A cumulative running total of each period's volume: added if the close is higher than the preceding close, subtracted if lower, and nothing if unchanged.



Our example, in million shares. Day 4's close is unchanged, so nothing is added.

Chart AM The OBV worked day by day.

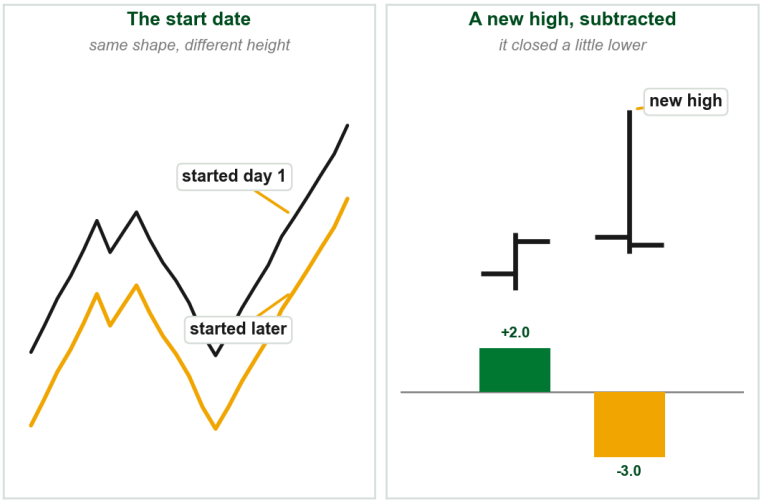
FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart AM lays the sum out in three panels: the closes, each day's volume signed by its close, and the running total.

The book names a disadvantage and a weakness. The disadvantage: the OBV's absolute value depends on the start date. Its relative changes do not, so all eight methods can still be used. The weakness: the whole of a period's volume is added or subtracted regardless of how the bar behaved. A bar that closes just below the previous close is subtracted in full, even if it made a new high far above the previous bar's high. To redress this, many practitioners prefer the ADL, the Chaikin Money Flow or the Chaikin Oscillator.

Chart AN draws both. On the left, the same OBV started on two days: two lines of the same shape at different heights. On the right, the second bar makes a much higher high but closes a little lower, and its whole 3.0 million is taken away.

Figure 6.39 shows Citigroup's price falling to a lower low while the OBV makes a higher low, bullish divergence; prices rebounded strongly. The OBV is price-based, so by Section 1's rule its bullish divergence is price falling while it rises.



Invented prices and volume. Right: the second bar's whole 3.0 million is taken away.

Chart AN The start date, and a new high subtracted.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

The daily chart of Citigroup with an On Balance Volume panel under it. A falling line marks price's lower lows in August to October 2011, and a rising line under the OBV, labelled bullish divergence, marks its higher lows.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 6.39 OBV bullish divergence on Citigroup.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

The daily chart of copper with an On Balance Volume panel under it. Two arrows mark price rising to higher peaks in 2011 while two arrows under the OBV fall, labelled bearish divergence.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 6.40 OBV bearish divergence on copper.

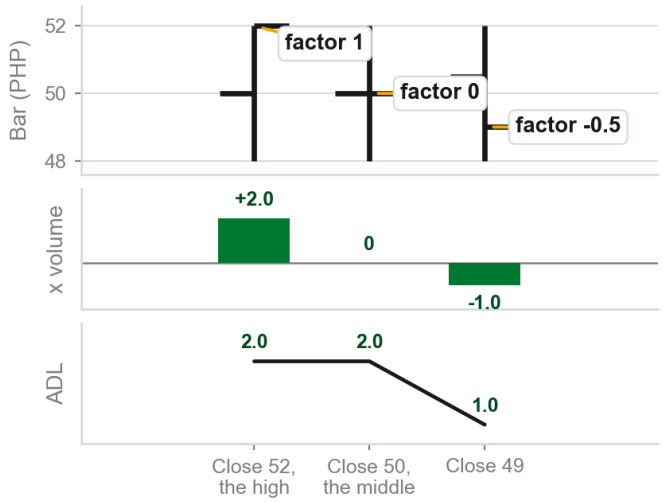
Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 6.40 is the mirror: copper's price rises twice while the OBV falls, bearish divergence, which reads the same on both kinds of oscillator. Prices fell strongly thereafter.

6.3 The Accumulation Distribution Line

The ADL also keeps a running total of volume, but each bar adds or subtracts only a share of its volume, set by where it closed in its range. C, H and L are the bar's close, high and low, from Chapter 1's OHLC. Take a bar with a high of 52 pesos and a low of 48, on 2 million shares. Closing at 52, its factor is 1 and it adds 2.0 million; at 50, the middle, the factor is 0 and it adds nothing; at 49 the factor is (1 minus 3) over 4, minus one half, and it subtracts 1.0 million.

> **Accumulation Distribution Line, ADL.** Calculate the multiplying factor [(C - L) - (H - C)] / (H - L) times the period's volume, and add it to its running total: the ADL.



Our example: each bar from PHP 48 to PHP 52 on 2 million shares.

Chart AO Three closes of one bar, worked through the ADL.

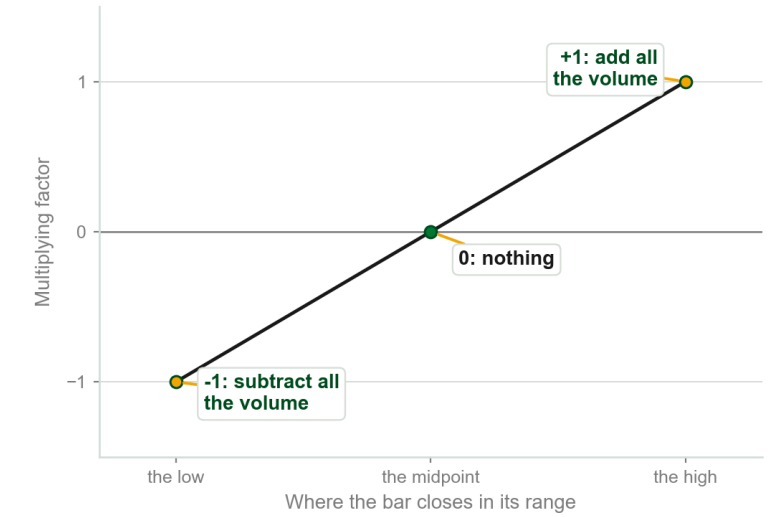
FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart AO works those three closes: the bar and its factor, the factor times the volume, and the running total, 2.0, 2.0 and 1.0 million. The name echoes the accumulation and distribution phases of Chapters 2 and 4; the book does not connect the two.

The algebra is the book's. If C minus L equals H minus C, then 2C equals H plus L and C is the midpoint, so a bar that closes at its midpoint adds nothing. The further above the midpoint it closes, the more volume is added, and the further below, the more subtracted. Putting the high in place of the close makes the factor exactly 1, so at most the day's total volume is added, never a multiple of it.

Chart AP draws the factor as one straight line: minus 1 when the bar closes at its low, 0 at its midpoint, plus 1 at its high. **The ADL accounts for the current bar's price action, which the OBV does not.**

Figure 6.41 is hourly EURUSD. The ADL rises with price, bullish confirmation, while the volume bars fall, bearish divergence: an underlying weakness in the uptrend. The book does not pick one; it says the trader should wait for price to confirm a reversal before taking part.



Not prices: the factor [(C - L) - (H - C)] / (H - L), for every close from low to high.

Chart AP The ADL's multiplying factor from the low to the high.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

An hourly EURUSD chart rising to a top, with an A/D line panel rising alongside it, labelled bullish confirmation, and a volume panel falling, labelled bearish divergence, under the heading divergence between volume and ADL.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 6.41 The ADL confirms while volume diverges on EURUSD.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

The daily chart of Exxon Mobil with OBV and ADL panels. On the left the two move apart, non-confirmation, under an erratic and volatile price; on the right they rise together, confirmation, under a pronounced trend.

ARTWORK NOT REPRODUCED IN THE PUBLIC BUILD

Figure 6.42 OBV and ADL together, trend and range, on Exxon Mobil.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

Figure 6.42 reads the OBV and the ADL together. Where they rise or fall together, price trends decisively; where they move in opposite directions, price tends to range or consolidate.

Compare the left ellipse, non-confirmation under an erratic price, with the right one, confirmation under a pronounced trend. The book says this helps gauge regime changes and decide between a trend-following and a range-bound strategy.

6.4 The Chaikin Money Flow and Oscillator

Both are built from the ADL's signed volume, S, the multiplying factor times the period's volume. N is a number of periods. Over three bars with signed volumes of 2.0, 0 and minus 0.5 million on volumes of 2.0, 1.0 and 1.0 million, the sum of S is 1.5 and the sum of volume 4.0, so the CMF is 0.375. The book then says to keep a running total of the CMF, as for the ADL, and to take the Chaikin oscillator from it as a shorter-term EMA less a longer-term EMA. An EMA is an exponential moving average, taught in the book's Chapter 11, and the book gives no lengths for the two.

> **Chaikin Money Flow and Chaikin Oscillator.** S is the multiplying factor times volume. N-period CMF: the N-period sum of S over the N-period sum of volume. Chaikin oscillator: a shorter-term EMA less a longer-term EMA of the CMF's running total.

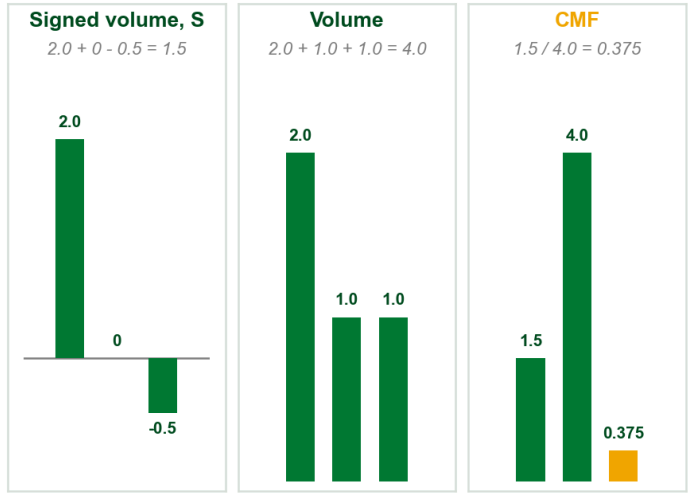


Chart AQ The Chaikin Money Flow worked for three bars.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart AQ sets the CMF out as three columns: the signed volumes and their sum, the volumes and their sum, and the one divided by the other.

In Figure 6.43 the Chaikin oscillator gives an earlier sell, or bearish, signal than the CMF: follow the dashed arrows from each panel up to the top, and the oscillator's comes first. Oversold levels on both point to a potential bottom in copper around October 2011. The figure prints settings of 20 for the CMF and 3 and 10 for the oscillator; the text gives none.

6.5 The NYSE Up/Down Volume, the Herrick index and the COT report

The NYSE Up/Down Volume sets the volume in the exchange's rising stocks against the volume in its falling ones. It is non-price-based, so its divergence is read like plain volume.

> **NYSE Up/Down Volume.** The ratio of advancing volume over declining volume. Being non-price-based, its divergence is read like volume bars and smoothed and detrended volume.



Figure 6.43 The Chaikin pair on copper.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

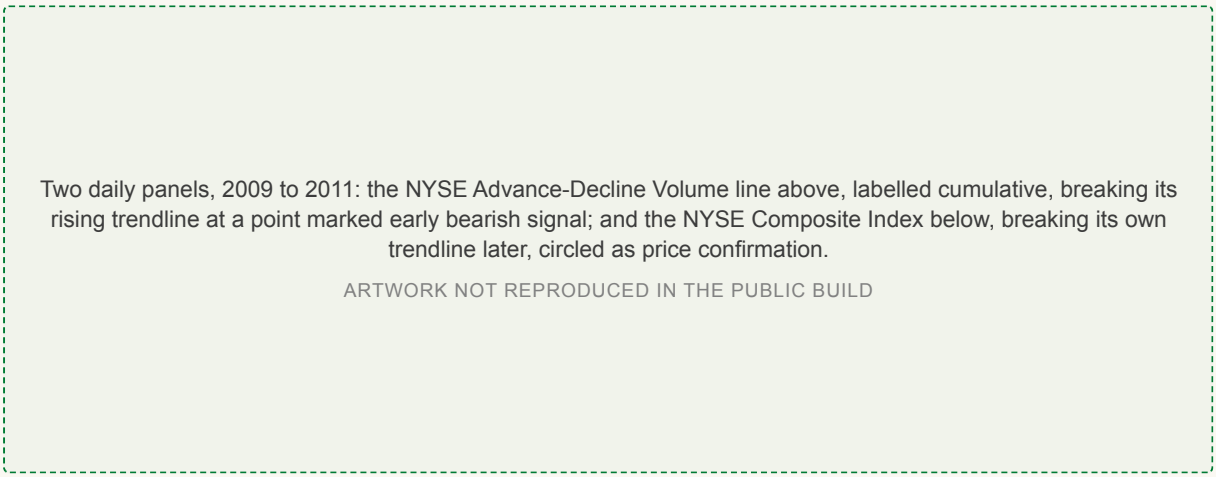


Figure 6.44 The NYSE Up/Down Volume line giving an early bearish signal.

Reproduced from Lim, The Handbook of Technical Analysis (Wiley, 2016)

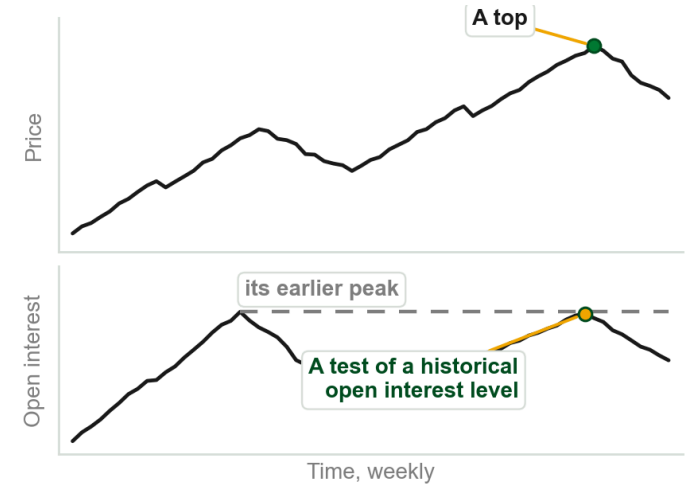
In Figure 6.44 the volume line breaks its rising trendline first, an early bearish signal, and the NYSE Composite Index breaks its own line later, the price confirmation. **The book says two things here: its text calls the indicator a ratio, while its figure labels the line cumulative and the line runs below zero, which a ratio cannot.**

The Herrick Payoff Index uses open interest in its calculations, so it can only be used with futures and options, which have open interest; a share on the PSE has none. It incorporates price, so its divergence is read as for any price-based volume oscillator. The book gives no formula and no figure for it.

> **Herrick Payoff Index.** The Herrick Payoff Index uses open interest in its calculations, so it may only be used with futures and options. It incorporates price, so its divergence is read as for all price-based volume oscillators. Bearish signals: money flowing out; bullish signals: money flowing in.

Last, the Commitment of Traders, COT, report, which says who holds the open positions behind a futures market's open interest. Hedgers were named among the market participants in Chapter 1. Traders and investors use the report to identify potential tops and bottoms, looking for changes in the net commercial or non-commercial open positions, and for a test of any significant or historical open interest level. The book does not define net.

Commitment of Traders, COT, report. Released every Friday, it reports all open positions held by commercials (hedgers, non-cash positions), non-commercials (hedge funds, financial institutions, large traders) and, in aggregate, small traders.



Invented prices and volume.
The COT report gives the positions behind this open interest.

Chart AR Open interest testing its earlier historical peak.

FIN1209 teaching chart, drawn for this course. Illustrative data, not a real instrument.

Chart AR draws the second of those uses: an invented futures price rising to a top while its open interest climbs back to the dashed line at its earlier peak, a test of a historical open interest level.

CHECK YOURSELF

A bar closes exactly at the midpoint of its range on 4 million shares. What does it add to the OBV if its close is higher than yesterday's, and what does it add to the ADL?

Chapter 6 in five sentences

Summary

1. Volume is participation: moving with price, bullish; against it, bearish.
2. Volume falling is divergence, rising is confirmation, whatever price does.
3. Minimum volume times breakouts and reversals; extreme volume, climaxes.
4. Barriers, filters and volume by price pick out the volume that matters.
5. Open interest reads like volume; the OBV, ADL and Chaikin pair use price.

Review questions

These are the chapter's own review questions. Every one is answered somewhere in the six sections above, and they are the shape the quiz takes.

1. Volume against tick volume? Bearish and bullish volume?
2. How is volume divergence distinct from standard oscillator divergence?
3. How would you use volume as a timing indicator?
4. The various modes of volume filtering?
5. The relationship between volume, open interest and price?
6. Absolute or relative volume? Volume cycles: their use, and where?

Key terms

Every term this chapter defines, against the subsection that defines it.

Volume	1.3
Bullish volume and bearish volume	1.4
Volume divergence and volume confirmation	1.5
Bullish and bearish divergence and confirmation	1.6
Price-based and non-price-based volume oscillators	1.9
Low volume breakout signal	2.1
Low volume reversal	2.1
Volume-weighted moving average, VWMA	2.2
Upper and lower overextended volume levels	2.3
Low volume top and low volume bottom	2.4
Smoothed volume filter	3.2
Two- or three-sigma volume filter	3.2
Volume filtered inflection points	3.3
Loss aversion bias	4.1
Open interest	4.2
Tick volume	4.3
The Arms Index, or TRIN	4.4
Volume by price and volume by time	4.5
Rest, reversal and continuation bars	5.1
Intraday volume cycle	5.4
On Balance Volume, OBV	6.2
Accumulation Distribution Line, ADL	6.3
Chaikin Money Flow and Chaikin Oscillator	6.4
NYSE Up/Down Volume	6.5
Herrick Payoff Index	6.5
Commitment of Traders, COT, report	6.5

Sources

- Lim, M. (2016). The Handbook of Technical Analysis. Wiley. Chapter 6, and the source of every figure here.